MHC Diversity

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Contents

Characterizing Diversity at MHC Loci

The major histocompatibility complex (MHC) is an important component of the immune response in vertebrates. Numerous previous studies have documented evidence of balancing selection acting on the region. Given its role in adaptive immunity, levels of genetic diversity within the region are especially important to assess in populations of conservation concern. With this in mind, the probe panel used in our project included several probes designed to target MHC loci. The following references were used during probe design for this project.

1. Goda et al. (2010) - MHC class II *DRB* genes in Ursidae, evidence for balancing selection on the locus, likely duplicated (3 alleles detected) in *U. americanus*.
2. Yasukochi et al. (2010) - successfully sequenced expressed portions of the MHC class II *DQB* gene in *U. thibetanus*, identified exon 2
3. Yasukochi et al. (2012) - assess diversity in exon 2 of *DQB* in *U. thibetanus japonicus*. Evidence for balancing selection on the locus, likely reduced diversity due to a bottleneck.
4. Kuduk et al. (2012) - surveyed diversity in MHC class I, *DRB*, *DQA*, *DQB* (latter 3 are MHC class II genes) in *U. arctos*. High allelic diversity at MHC class I and *DRB*. The rate of non-synonymous substitution (d_N) was higher than the rate of synonymous substitution (d_S) at antigen binding sites in *DRB* and *DQB* (evidence for selection).
5. Weber et al. (2013) - low diversity at *DQA* and *DQB* in *U. maritimus*.
6. Ishibashi et al. (2017) - further loss of diversity in exon 2 of *DQB* in *U. thibetanus japonicus*, compared to Yasukochi et al. (2012).
7. Jia et al. (2023) - focused on exon 2 of *DQB* in *U. thibetanus*, high allelic diversity, evidence for trans-species polymorphism. $d_N > d_S$ at codons in peptide-binding regions.

Sequence accessions from these studies were used to develop capture-seq probes that targeted MHC genes.

Looking for targeted MHC markers

We started by searching for the accession numbers identified above in the *Ursus americanus* reference genome (using "BLAST the reference genome" from the NCBI taxonomy page for *U. americanus*). We then looked for matching regions from the reference genome in the raw VCF from RAPiD Genomics using `grep` to pull out the relevant scaffold (`CHROM` column in the VCF). Results of these searches can be reviewed in this spreadsheet (see "Second Attempt" tab for the most recent set of results).

Only two of the regions identified by BLAST were present in our raw VCF:

1. `NW_025577149.1:1531301-1531698` - identified by several studies, annotated in the reference genome as *DQB*, looking more closely with `snpEff` (Cingolani et al. 2012), we can narrow exon 2 down to `NW_025577149.1:1531458-1531646`.
2. `NW_025577149.1:1535166-1535535` - close match to a portion of the cDNA sequence from Yasukochi et al. (2010), appears to be exon 6 of *DQB* according to the NCBI genome data viewer. Looking more closely with `snpEff` (Cingolani et al. 2012), most of this chunk is 3' UTR.

We'll extract the coding sequences from these windows and summarize diversity in each of the populations at SNPs within these regions (including the number of non-synonymous substitutions, as inferred by the `snpEff` annotation).

Extracting MHC markers

This chunk of code creates `.bed` files for extracting the markers around exons 2 and 6 with `vcftools`. We'll start with the `PopGen0.recode.vcf` file that served as the beginning point for our filtering in `1_filterPopGen`. This file has had duplicate SNPs, half-calls, indels, and duplicate genotypes removed, but no other filters applied. We also want to drop samples with high levels of excess heterozygosity, so we'll retain only samples present in `PopGen.vcf`. First, we extract SNPs within these regions to identify boundaries of the translated sequence, later we will apply some filters for SNP quality and genotype depth.

```
grep ^#CHROM ../snpfiltering/PopGen.vcf | tr '\t' '\n' | grep ^UGA > keepUs.txt
#DQB exon 2: NW_025577149.1:1531301-1531698
echo 'CHROM START END' > DQBex2.bed
echo 'NW_025577149.1 1531300 1531699' >> DQBex2.bed
vcftools --vcf ../snpfiltering/PopGen0.recode.vcf \
          --bed DQBex2.bed --keep keepUs.txt --out DQBex2 --recode

#DQB exon 6: NW_025577149.1:1535166-1535535
echo 'CHROM START END' > DQBex6.bed
echo 'NW_025577149.1 1535165 1535536' >> DQBex6.bed
vcftools --vcf ../snpfiltering/PopGen0.recode.vcf \
          --bed DQBex6.bed --keep keepUs.txt --out DQBex6 --recode
```

```
VCFtools - 0.1.17
(C) Adam Auton and Anthony Marcketta 2009

Parameters as interpreted:
    --vcf ../snpfiltering/PopGen0.recode.vcf
    --keep keepUs.txt
    --out DQBex2
    --recode
    --bed DQBex2.bed

Warning: Expected at least 2 parts in INFO entry: ID=AF,Number=A,Type=Float,Description="Estimated allel
Warning: Expected at least 2 parts in INFO entry: ID=PRO,Number=1,Type=Float,Description="Reference all
Warning: Expected at least 2 parts in INFO entry: ID=PAO,Number=A,Type=Float,Description="Alternate all
Warning: Expected at least 2 parts in INFO entry: ID=SRP,Number=1,Type=Float,Description="Strand balanc
Warning: Expected at least 2 parts in INFO entry: ID=SAP,Number=A,Type=Float,Description="Strand balanc
Warning: Expected at least 2 parts in INFO entry: ID=AB,Number=A,Type=Float,Description="Allele balance
Warning: Expected at least 2 parts in INFO entry: ID=ABP,Number=A,Type=Float,Description="Allele balanc
Warning: Expected at least 2 parts in INFO entry: ID=RPP,Number=A,Type=Float,Description="Read Placemen
Warning: Expected at least 2 parts in INFO entry: ID=RPPR,Number=1,Type=Float,Description="Read Placeme
Warning: Expected at least 2 parts in INFO entry: ID=EPP,Number=A,Type=Float,Description="End Placement
Warning: Expected at least 2 parts in INFO entry: ID=EPPR,Number=1,Type=Float,Description="End Placemen
Warning: Expected at least 2 parts in INFO entry: ID=TYPE,Number=A,Type=String,Description="The type of
Warning: Expected at least 2 parts in INFO entry: ID=TYPE,Number=A,Type=String,Description="The type of
Warning: Expected at least 2 parts in INFO entry: ID=TYPE,Number=A,Type=String,Description="The type of
Warning: Expected at least 2 parts in INFO entry: ID=TYPE,Number=A,Type=String,Description="The type of
Warning: Expected at least 2 parts in INFO entry: ID=TYPE,Number=A,Type=String,Description="The type of
Warning: Expected at least 2 parts in INFO entry: ID=CIGAR,Number=A,Type=String,Description="The extend
Warning: Expected at least 2 parts in FORMAT entry: ID=GQ,Number=1,Type=Float,Description="Genotype Qua
Warning: Expected at least 2 parts in FORMAT entry: ID=GL,Number=G,Type=Float,Description="Genotype Lik
Keeping individuals in 'keep' list
```

```
After filtering, kept 260 out of 268 Individuals
Outputting VCF file...
    Read 2 BED file entries.
After filtering, kept 144 out of a possible 312565 Sites
Run Time = 5.00 seconds

VCFtools - 0.1.17
(C) Adam Auton and Anthony Marcketta 2009

Parameters as interpreted:
    --vcf ../snpfiltering/PopGen0.recode.vcf
    --keep keepUs.txt
    --out DQBex6
    --recode
    --bed DQBex6.bed

Warning: Expected at least 2 parts in INFO entry: ID=AF,Number=A,Type=Float,Description="Estimated allel
Warning: Expected at least 2 parts in INFO entry: ID=PRO,Number=1,Type=Float,Description="Reference allel
Warning: Expected at least 2 parts in INFO entry: ID=PAO,Number=A,Type=Float,Description="Alternate allel
Warning: Expected at least 2 parts in INFO entry: ID=SRP,Number=1,Type=Float,Description="Strand balance
Warning: Expected at least 2 parts in INFO entry: ID=SAP,Number=A,Type=Float,Description="Strand balance
Warning: Expected at least 2 parts in INFO entry: ID=AB,Number=A,Type=Float,Description="Allele balance
Warning: Expected at least 2 parts in INFO entry: ID=ABP,Number=A,Type=Float,Description="Allele balance
Warning: Expected at least 2 parts in INFO entry: ID=RPP,Number=A,Type=Float,Description="Read Placement
Warning: Expected at least 2 parts in INFO entry: ID=RPPR,Number=1,Type=Float,Description="Read Placemen
Warning: Expected at least 2 parts in INFO entry: ID=EPP,Number=A,Type=Float,Description="End Placement
Warning: Expected at least 2 parts in INFO entry: ID=EPPR,Number=1,Type=Float,Description="End Placement
Warning: Expected at least 2 parts in INFO entry: ID=TYPE,Number=A,Type=String,Description="The type of
Warning: Expected at least 2 parts in INFO entry: ID=TYPE,Number=A,Type=String,Description="The type of
Warning: Expected at least 2 parts in INFO entry: ID=TYPE,Number=A,Type=String,Description="The type of
Warning: Expected at least 2 parts in INFO entry: ID=TYPE,Number=A,Type=String,Description="The type of
Warning: Expected at least 2 parts in INFO entry: ID=TYPE,Number=A,Type=String,Description="The type of
Warning: Expected at least 2 parts in INFO entry: ID=CIGAR,Number=A,Type=String,Description="The extende
Warning: Expected at least 2 parts in FORMAT entry: ID=GQ,Number=1,Type=Float,Description="Genotype Qual
Warning: Expected at least 2 parts in FORMAT entry: ID=GL,Number=G,Type=Float,Description="Genotype Like
Keeping individuals in 'keep' list
After filtering, kept 260 out of 268 Individuals
Outputting VCF file...
    Read 2 BED file entries.
After filtering, kept 76 out of a possible 312565 Sites
Run Time = 3.00 seconds
```

Annotate SNPs in these regions

We will annotate the SNPs using `snpEff` (Cingolani et al. 2012), downloaded from this page or (in the case of our virtual environment) installed via `conda` from the `bioconda` channel. In order to use this software to annotate SNPs in our vcf, we first need to configure a new genome in the `snpEff` database. We followed instructions from the `snpEff` GitHub page for these steps.

n.b., the reference genome configuration was completed in an interactive session before creating this markdown (`eval=FALSE` for the chunks below).

`snpEff` setup

Download reference genome

First, we downloaded the reference genome files from NCBI's *U. americanus* taxonomy page and renamed them as follows:

i. `sequences.fa` - the reference genome in FASTA format ("Genome sequences" from NCBI download)
ii. `genes.gtf` - annotation information in GTF format ("Annotation features (GTF)" from NCBI download)

These files were saved into a `data/jax_bbear` subdirectory in our virtual environment:
`veGAbears/share/snpeff-5.2-1/data/jax_bbear`

Configure the `snpEff` database

We then edit the configuration file (`veGAbears/share/snpeff-5.2-1/snpEff.config`) to include the following lines (in the `Third party databases` section of the `.config` file):

```
# Black bear genome, jax lab
jax_bbear.genome : Black Bear
```

After that, we create the new database using `snpEff` (*n.b.*, `eval=FALSE` for this chunk). Note that we are skipping two checks that can be performed by `snpEff` with the `-noCheckCds` and `-noCheckProtein` flags.

```
snpEff build -gtf22 -v jax_bbear -noCheckCds -noCheckProtein
```

Test configuration

The step above takes a few minutes (~7 min, when run interactively), but prints lots of information on progress. Once it is complete, we can check that the genome was configured successfully with...

```
snpEff dump jax_bbear | head
```

```
#-----------------------------------------------
# Genome name                 : 'Black Bear'
# Genome version              : 'jax_bbear'
# Genome ID                   : 'jax_bbear[0]'
# Has protein coding info     : true
# Has Tr. Support Level info : true
# Genes                       : 28921
# Protein coding genes        : 21094
#-----------------------------------------------
# Transcripts                 : 54636
```

Annotate SNPs from *DQB*

Now annotate the two *DQB* exons with `snpEff`. This will help us identify the boundaries of the exon and avoid splice region variants. It will also give us information on the nature of the mutations underlying each SNP (*i.e.*, missense, nonsense, etc.).

Exon 2

During annotation, `snpEff` adds a line describing the information in the annotation field to the header of the `.vcf` file:

```
snpEff jax_bbear DQBex2.recode.vcf > DQBex2.ann.vcf
grep ANN DQBex2.ann.vcf | head -1
```

```
##INFO=<ID=ANN,Number=.,Type=String,Description="Functional annotations: 'Allele | Annotation | Annotation_Impact | Gene_Name | Gene_ID | Feature_Type | Feature_ID | Transcript_BioType | Rank | HGVS.c | H
```

Here are the annotations for exon 2:

```
grep -v ^# DQBex2.ann.vcf | cut -f 1,2,8
```

```
NW_025577149.1  1531304 ANN=T|intron_variant|MODIFIER|LOC123790950|LOC123790950|transcript|XM_045795559.1|protein_coding|1/5|c.98-92G>T||||||
NW_025577149.1  1531316 ANN=A|intron_variant|MODIFIER|LOC123790950|LOC123790950|transcript|XM_045795559.1|protein_coding|1/5|c.98-80C>A||||||
NW_025577149.1  1531323 ANN=T|intron_variant|MODIFIER|LOC123790950|LOC123790950|transcript|XM_045795559.1|protein_coding|1/5|c.98-73G>T||||||
NW_025577149.1  1531325 ANN=A|intron_variant|MODIFIER|LOC123790950|LOC123790950|transcript|XM_045795559.1|protein_coding|1/5|c.98-71C>A||||||
NW_025577149.1  1531329 ANN=T|intron_variant|MODIFIER|LOC123790950|LOC123790950|transcript|XM_045795559.1|protein_coding|1/5|c.98-67C>T||||||
NW_025577149.1  1531330 ANN=T|intron_variant|MODIFIER|LOC123790950|LOC123790950|transcript|XM_045795559.1|protein_coding|1/5|c.98-66C>T||||||
NW_025577149.1  1531341 ANN=T|intron_variant|MODIFIER|LOC123790950|LOC123790950|transcript|XM_045795559.1|protein_coding|1/5|c.98-55C>T||||||
NW_025577149.1  1531345 ANN=T|intron_variant|MODIFIER|LOC123790950|LOC123790950|transcript|XM_045795559.1|protein_coding|1/5|c.98-51G>T||||||
NW_025577149.1  1531347 ANN=T|intron_variant|MODIFIER|LOC123790950|LOC123790950|transcript|XM_045795559.1|protein_coding|1/5|c.98-49G>T||||||
NW_025577149.1  1531355 ANN=T|intron_variant|MODIFIER|LOC123790950|LOC123790950|transcript|XM_045795559.1|protein_coding|1/5|c.98-41G>T||||||
NW_025577149.1  1531358 ANN=A|intron_variant|MODIFIER|LOC123790950|LOC123790950|transcript|XM_045795559.1|protein_coding|1/5|c.98-38G>A||||||
NW_025577149.1  1531359 ANN=G|intron_variant|MODIFIER|LOC123790950|LOC123790950|transcript|XM_045795559.1|protein_coding|1/5|c.98-37C>G||||||
NW_025577149.1  1531361 ANN=T|intron_variant|MODIFIER|LOC123790950|LOC123790950|transcript|XM_045795559.1|protein_coding|1/5|c.98-35C>T||||||
NW_025577149.1  1531362 ANN=T|intron_variant|MODIFIER|LOC123790950|LOC123790950|transcript|XM_045795559.1|protein_coding|1/5|c.98-34G>T||||||
NW_025577149.1  1531363 ANN=T|intron_variant|MODIFIER|LOC123790950|LOC123790950|transcript|XM_045795559.1|protein_coding|1/5|c.98-33G>T||||||
NW_025577149.1  1531366 ANN=G|intron_variant|MODIFIER|LOC123790950|LOC123790950|transcript|XM_045795559.1|protein_coding|1/5|c.98-30A>G||||||
NW_025577149.1  1531378 ANN=A|intron_variant|MODIFIER|LOC123790950|LOC123790950|transcript|XM_045795559.1|protein_coding|1/5|c.98-18C>A||||||
NW_025577149.1  1531379 ANN=A|intron_variant|MODIFIER|LOC123790950|LOC123790950|transcript|XM_045795559.1|protein_coding|1/5|c.98-17G>A||||||
NW_025577149.1  1531380 ANN=T|intron_variant|MODIFIER|LOC123790950|LOC123790950|transcript|XM_045795559.1|protein_coding|1/5|c.98-16C>T||||||
NW_025577149.1  1531381 ANN=C|intron_variant|MODIFIER|LOC123790950|LOC123790950|transcript|XM_045795559.1|protein_coding|1/5|c.98-15G>C||||||
NW_025577149.1  1531384 ANN=T|intron_variant|MODIFIER|LOC123790950|LOC123790950|transcript|XM_045795559.1|protein_coding|1/5|c.98-12G>T||||||
NW_025577149.1  1531386 ANN=G|intron_variant|MODIFIER|LOC123790950|LOC123790950|transcript|XM_045795559.1|protein_coding|1/5|c.98-10T>G||||||
NW_025577149.1  1531387 ANN=C|intron_variant|MODIFIER|LOC123790950|LOC123790950|transcript|XM_045795559.1|protein_coding|1/5|c.98-9T>C||||||
NW_025577149.1  1531389 ANN=T|splice_region_variant&intron_variant|LOW|LOC123790950|LOC123790950|transcript|XM_045795559.1|protein_coding|1/5|c.98-7C>T||||||
NW_025577149.1  1531392 ANN=A|splice_region_variant&intron_variant|LOW|LOC123790950|LOC123790950|transcript|XM_045795559.1|protein_coding|1/5|c.98-4G>A||||||
NW_025577149.1  1531393 ANN=A|splice_region_variant&intron_variant|LOW|LOC123790950|LOC123790950|transcript|XM_045795559.1|protein_coding|1/5|c.98-3C>A||||||
NW_025577149.1  1531396 ANN=G|missense_variant&splice_region_variant|MODERATE|LOC123790950|LOC123790950|transcript|XM_045795559.1|protein_coding|2/6|c.98A>G|p.Gln33Arg|210/1262|98/798|33/265||
NW_025577149.1  1531398 ANN=T|missense_variant&splice_region_variant|MODERATE|LOC123790950|LOC123790950|transcript|XM_045795559.1|protein_coding|2/6|c.100G>T|p.Asp34Tyr|212/1262|100/798|34/265||
NW_025577149.1  1531403 ANN=T|synonymous_variant|LOW|LOC123790950|LOC123790950|transcript|XM_045795559.1|protein_coding|2/6|c.105C>T|p.Phe35Phe|217/1262|105/798|35/265||
NW_025577149.1  1531404 ANN=T|missense_variant|MODERATE|LOC123790950|LOC123790950|transcript|XM_045795559.1|protein_coding|2/6|c.106G>T|p.Val36Leu|218/1262|106/798|36/265||
NW_025577149.1  1531406 ANN=T|synonymous_variant|LOW|LOC123790950|LOC123790950|transcript|XM_045795559.1|protein_coding|2/6|c.108G>T|p.Val36Val|220/1262|108/798|36/265||
NW_025577149.1  1531407 ANN=G|missense_variant|MODERATE|LOC123790950|LOC123790950|transcript|XM_045795559.1|protein_coding|2/6|c.109C>G|p.Leu37Val|221/1262|109/798|37/265||,T|missense_variant|MODERATE|LOC
NW_025577149.1  1531408 ANN=A|missense_variant|MODERATE|LOC123790950|LOC123790950|transcript|XM_045795559.1|protein_coding|2/6|c.110T>A|p.Leu37His|222/1262|110/798|37/265||
NW_025577149.1  1531409 ANN=A|synonymous_variant|LOW|LOC123790950|LOC123790950|transcript|XM_045795559.1|protein_coding|2/6|c.111C>A|p.Leu37Leu|223/1262|111/798|37/265||
NW_025577149.1  1531415 ANN=C|synonymous_variant|LOW|LOC123790950|LOC123790950|transcript|XM_045795559.1|protein_coding|2/6|c.117T>C|p.Phe39Phe|229/1262|117/798|39/265||
NW_025577149.1  1531418 ANN=T|missense_variant|MODERATE|LOC123790950|LOC123790950|transcript|XM_045795559.1|protein_coding|2/6|c.120G>T|p.Lys40Asn|232/1262|120/798|40/265||
NW_025577149.1  1531420 ANN=A|missense_variant|MODERATE|LOC123790950|LOC123790950|transcript|XM_045795559.1|protein_coding|2/6|c.122G>A|p.Gly41Asp|234/1262|122/798|41/265||,C|missense_variant|MODERATE|LOC
NW_025577149.1  1531421 ANN=A|synonymous_variant|LOW|LOC123790950|LOC123790950|transcript|XM_045795559.1|protein_coding|2/6|c.123C>A|p.Gly41Gly|235/1262|123/798|41/265||,T|synonymous_variant|LOW|LOC123790
NW_025577149.1  1531422 ANN=A|missense_variant|MODERATE|LOC123790950|LOC123790950|transcript|XM_045795559.1|protein_coding|2/6|c.124G>A|p.Glu42Lys|236/1262|124/798|42/265||,C|missense_variant|MODERATE|LOC
NW_025577149.1  1531423 ANN=T|missense_variant|MODERATE|LOC123790950|LOC123790950|transcript|XM_045795559.1|protein_coding|2/6|c.125A>T|p.Glu42Val|237/1262|125/798|42/265||
NW_025577149.1  1531426 ANN=T|missense_variant|MODERATE|LOC123790950|LOC123790950|transcript|XM_045795559.1|protein_coding|2/6|c.128G>T|p.Cys43Phe|240/1262|128/798|43/265||
NW_025577149.1  1531430 ANN=T|synonymous_variant|LOW|LOC123790950|LOC123790950|transcript|XM_045795559.1|protein_coding|2/6|c.132C>T|p.Tyr44Tyr|244/1262|132/798|44/265||
NW_025577149.1  1531433 ANN=A|missense_variant|MODERATE|LOC123790950|LOC123790950|transcript|XM_045795559.1|protein_coding|2/6|c.135C>A|p.Phe45Leu|247/1262|135/798|45/265||
NW_025577149.1  1531436 ANN=A|synonymous_variant|LOW|LOC123790950|LOC123790950|transcript|XM_045795559.1|protein_coding|2/6|c.138C>A|p.Thr46Thr|250/1262|138/798|46/265||
NW_025577149.1  1531439 ANN=T|synonymous_variant|LOW|LOC123790950|LOC123790950|transcript|XM_045795559.1|protein_coding|2/6|c.141C>T|p.Asn47Asn|253/1262|141/798|47/265||
NW_025577149.1  1531440 ANN=T|missense_variant|MODERATE|LOC123790950|LOC123790950|transcript|XM_045795559.1|protein_coding|2/6|c.142G>T|p.Gly48Trp|254/1262|142/798|48/265||
NW_025577149.1  1531441 ANN=T|missense_variant|MODERATE|LOC123790950|LOC123790950|transcript|XM_045795559.1|protein_coding|2/6|c.143G>T|p.Gly48Val|255/1262|143/798|48/265||
NW_025577149.1  1531442 ANN=A|synonymous_variant|LOW|LOC123790950|LOC123790950|transcript|XM_045795559.1|protein_coding|2/6|c.144G>A|p.Gly48Gly|256/1262|144/798|48/265||
NW_025577149.1  1531445 ANN=C|synonymous_variant|LOW|LOC123790950|LOC123790950|transcript|XM_045795559.1|protein_coding|2/6|c.147A>C|p.Thr49Thr|259/1262|147/798|49/265||,G|synonymous_variant|LOW|LOC123790
NW_025577149.1  1531446 ANN=C|missense_variant|MODERATE|LOC123790950|LOC123790950|transcript|XM_045795559.1|protein_coding|2/6|c.148G>C|p.Glu50Gln|260/1262|148/798|50/265||
NW_025577149.1  1531450 ANN=T|missense_variant|MODERATE|LOC123790950|LOC123790950|transcript|XM_045795559.1|protein_coding|2/6|c.152G>T|p.Arg51Leu|264/1262|152/798|51/265||
NW_025577149.1  1531451 ANN=C|synonymous_variant|LOW|LOC123790950|LOC123790950|transcript|XM_045795559.1|protein_coding|2/6|c.153G>C|p.Arg51Arg|265/1262|153/798|51/265||
NW_025577149.1  1531454 ANN=A|synonymous_variant|LOW|LOC123790950|LOC123790950|transcript|XM_045795559.1|protein_coding|2/6|c.156G>A|p.Val52Val|268/1262|156/798|52/265||
NW_025577149.1  1531458 ANN=C|missense_variant|MODERATE|LOC123790950|LOC123790950|transcript|XM_045795559.1|protein_coding|2/6|c.160G>C|p.Gly54Arg|272/1262|160/798|54/265||
NW_025577149.1  1531459 ANN=T|missense_variant|MODERATE|LOC123790950|LOC123790950|transcript|XM_045795559.1|protein_coding|2/6|c.161G>T|p.Gly54Val|273/1262|161/798|54/265||
NW_025577149.1  1531464 ANN=G|missense_variant|MODERATE|LOC123790950|LOC123790950|transcript|XM_045795559.1|protein_coding|2/6|c.166A>G|p.Thr56Ala|278/1262|166/798|56/265||
NW_025577149.1  1531465 ANN=A|missense_variant|MODERATE|LOC123790950|LOC123790950|transcript|XM_045795559.1|protein_coding|2/6|c.167C>A|p.Thr56Asn|279/1262|167/798|56/265||
NW_025577149.1  1531470 ANN=T|missense_variant|MODERATE|LOC123790950|LOC123790950|transcript|XM_045795559.1|protein_coding|2/6|c.172C>T|p.His58Tyr|284/1262|172/798|58/265||
NW_025577149.1  1531471 ANN=C|missense_variant|MODERATE|LOC123790950|LOC123790950|transcript|XM_045795559.1|protein_coding|2/6|c.173A>C|p.His58Pro|285/1262|173/798|58/265||
NW_025577149.1  1531473 ANN=G|missense_variant|MODERATE|LOC123790950|LOC123790950|transcript|XM_045795559.1|protein_coding|2/6|c.175A>G|p.Ile59Val|287/1262|175/798|59/265||
NW_025577149.1  1531478 ANN=C|synonymous_variant|LOW|LOC123790950|LOC123790950|transcript|XM_045795559.1|protein_coding|2/6|c.180T>C|p.Tyr60Tyr|292/1262|180/798|60/265||
NW_025577149.1  1531482 ANN=A|synonymous_variant|LOW|LOC123790950|LOC123790950|transcript|XM_045795559.1|protein_coding|2/6|c.184C>A|p.Arg62Arg|296/1262|184/798|62/265||
NW_025577149.1  1531483 ANN=A|missense_variant|MODERATE|LOC123790950|LOC123790950|transcript|XM_045795559.1|protein_coding|2/6|c.185G>A|p.Arg62Gln|297/1262|185/798|62/265||,T|missense_variant|MODERATE|LOC
NW_025577149.1  1531484 ANN=A|synonymous_variant|LOW|LOC123790950|LOC123790950|transcript|XM_045795559.1|protein_coding|2/6|c.186G>A|p.Arg62Arg|298/1262|186/798|62/265||,C|synonymous_variant|LOW|LOC123790
NW_025577149.1  1531485 ANN=T|stop_gained|HIGH|LOC123790950|LOC123790950|transcript|XM_045795559.1|protein_coding|2/6|c.187G>T|p.Glu63*|299/1262|187/798|63/265||;LOF=(LOC123790950|LOC123790950|1|1.00);NMD
NW_025577149.1  1531487 ANN=T|missense_variant|MODERATE|LOC123790950|LOC123790950|transcript|XM_045795559.1|protein_coding|2/6|c.189G>T|p.Glu63Asp|301/1262|189/798|63/265||,A|synonymous_variant|LOW|LOC123
NW_025577149.1  1531488 ANN=T|stop_gained|HIGH|LOC123790950|LOC123790950|transcript|XM_045795559.1|protein_coding|2/6|c.190G>T|p.Glu64*|302/1262|190/798|64/265||;LOF=(LOC123790950|LOC123790950|1|1.00);NMD
NW_025577149.1  1531491 ANN=A|missense_variant|MODERATE|LOC123790950|LOC123790950|transcript|XM_045795559.1|protein_coding|2/6|c.193T>A|p.Phe65Ile|305/1262|193/798|65/265||
NW_025577149.1  1531492 ANN=A|missense_variant|MODERATE|LOC123790950|LOC123790950|transcript|XM_045795559.1|protein_coding|2/6|c.194T>A|p.Phe65Tyr|306/1262|194/798|65/265||
NW_025577149.1  1531495 ANN=C|missense_variant|MODERATE|LOC123790950|LOC123790950|transcript|XM_045795559.1|protein_coding|2/6|c.197T>C|p.Val66Ala|309/1262|197/798|66/265||,G|missense_variant|MODERATE|LOC
NW_025577149.1  1531499 ANN=A|synonymous_variant|LOW|LOC123790950|LOC123790950|transcript|XM_045795559.1|protein_coding|2/6|c.201C>A|p.Arg67Arg|313/1262|201/798|67/265||
NW_025577149.1  1531501 ANN=A|missense_variant|MODERATE|LOC123790950|LOC123790950|transcript|XM_045795559.1|protein_coding|2/6|c.203T>A|p.Phe68Tyr|315/1262|203/798|68/265||
NW_025577149.1  1531502 ANN=T|synonymous_variant|LOW|LOC123790950|LOC123790950|transcript|XM_045795559.1|protein_coding|2/6|c.204C>T|p.Phe68Phe|316/1262|204/798|68/265||
NW_025577149.1  1531503 ANN=T|missense_variant|MODERATE|LOC123790950|LOC123790950|transcript|XM_045795559.1|protein_coding|2/6|c.205G>T|p.Asp69Tyr|317/1262|205/798|69/265||
NW_025577149.1  1531506 ANN=G|missense_variant|MODERATE|LOC123790950|LOC123790950|transcript|XM_045795559.1|protein_coding|2/6|c.208A>G|p.Ser70Gly|320/1262|208/798|70/265||
NW_025577149.1  1531507 ANN=A|missense_variant|MODERATE|LOC123790950|LOC123790950|transcript|XM_045795559.1|protein_coding|2/6|c.209G>A|p.Ser70Asn|321/1262|209/798|70/265||
NW_025577149.1  1531509 ANN=A|missense_variant|MODERATE|LOC123790950|LOC123790950|transcript|XM_045795559.1|protein_coding|2/6|c.211G>A|p.Asp71Asn|323/1262|211/798|71/265||
NW_025577149.1  1531512 ANN=A|missense_variant|MODERATE|LOC123790950|LOC123790950|transcript|XM_045795559.1|protein_coding|2/6|c.214G>A|p.Val72Met|326/1262|214/798|72/265||
NW_025577149.1  1531514 ANN=T|synonymous_variant|LOW|LOC123790950|LOC123790950|transcript|XM_045795559.1|protein_coding|2/6|c.216G>T|p.Val72Val|328/1262|216/798|72/265||
NW_025577149.1  1531516 ANN=A|missense_variant|MODERATE|LOC123790950|LOC123790950|transcript|XM_045795559.1|protein_coding|2/6|c.218G>A|p.Gly73Glu|330/1262|218/798|73/265||,T|missense_variant|MODERATE|LOC
NW_025577149.1  1531517 ANN=T|synonymous_variant|LOW|LOC123790950|LOC123790950|transcript|XM_045795559.1|protein_coding|2/6|c.219G>T|p.Gly73Gly|331/1262|219/798|73/265||
NW_025577149.1  1531518 ANN=T|stop_gained|HIGH|LOC123790950|LOC123790950|transcript|XM_045795559.1|protein_coding|2/6|c.220G>T|p.Glu74*|332/1262|220/798|74/265||;LOF=(LOC123790950|LOC123790950|1|1.00);NMD
NW_025577149.1  1531521 ANN=T|missense_variant|MODERATE|LOC123790950|LOC123790950|transcript|XM_045795559.1|protein_coding|2/6|c.223C>T|p.His75Tyr|335/1262|223/798|75/265||
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NW_025577149.1  1531525 ANN=A|missense_variant|MODERATE|LOC123790950|LOC123790950|transcript|XM_045795559.1|protein_coding|2/6|c.227G>A|p.Arg76Gln|339/1262|227/798|76/265||
NW_025577149.1  1531526 ANN=C|synonymous_variant|LOW|LOC123790950|LOC123790950|transcript|XM_045795559.1|protein_coding|2/6|c.228G>C|p.Arg76Arg|340/1262|228/798|76/265||,T|synonymous_variant|LOW|LOC123790
NW_025577149.1  1531527 ANN=C|missense_variant|MODERATE|LOC123790950|LOC123790950|transcript|XM_045795559.1|protein_coding|2/6|c.229G>C|p.Ala77Pro|341/1262|229/798|77/265||
NW_025577149.1  1531532 ANN=C|synonymous_variant|LOW|LOC123790950|LOC123790950|transcript|XM_045795559.1|protein_coding|2/6|c.234G>C|p.Val78Val|346/1262|234/798|78/265||
NW_025577149.1  1531535 ANN=C|synonymous_variant|LOW|LOC123790950|LOC123790950|transcript|XM_045795559.1|protein_coding|2/6|c.237G>C|p.Thr79Thr|349/1262|237/798|79/265||
NW_025577149.1  1531541 ANN=C|synonymous_variant|LOW|LOC123790950|LOC123790950|transcript|XM_045795559.1|protein_coding|2/6|c.243G>C|p.Leu81Leu|355/1262|243/798|81/265||,T|synonymous_variant|LOW|LOC123790
NW_025577149.1  1531544 ANN=A|synonymous_variant|LOW|LOC123790950|LOC123790950|transcript|XM_045795559.1|protein_coding|2/6|c.246G>A|p.Gly82Gly|358/1262|246/798|82/265||
NW_025577149.1  1531546 ANN=T|missense_variant|MODERATE|LOC123790950|LOC123790950|transcript|XM_045795559.1|protein_coding|2/6|c.248G>T|p.Arg83Leu|360/1262|248/798|83/265||
NW_025577149.1  1531547 ANN=T|synonymous_variant|LOW|LOC123790950|LOC123790950|transcript|XM_045795559.1|protein_coding|2/6|c.249G>T|p.Arg83Arg|361/1262|249/798|83/265||
NW_025577149.1  1531548 ANN=A|missense_variant|MODERATE|LOC123790950|LOC123790950|transcript|XM_045795559.1|protein_coding|2/6|c.250C>A|p.Pro84Thr|362/1262|250/798|84/265||,T|missense_variant|MODERATE|LOC
NW_025577149.1  1531549 ANN=A|missense_variant|MODERATE|LOC123790950|LOC123790950|transcript|XM_045795559.1|protein_coding|2/6|c.251C>A|p.Pro84His|363/1262|251/798|84/265||,G|missense_variant|MODERATE|LOC
NW_025577149.1  1531550 ANN=G|synonymous_variant|LOW|LOC123790950|LOC123790950|transcript|XM_045795559.1|protein_coding|2/6|c.252C>G|p.Pro84Pro|364/1262|252/798|84/265||
NW_025577149.1  1531552 ANN=A|missense_variant|MODERATE|LOC123790950|LOC123790950|transcript|XM_045795559.1|protein_coding|2/6|c.254T>A|p.Ile85Asn|366/1262|254/798|85/265||,C|missense_variant|MODERATE|LOC
NW_025577149.1  1531553 ANN=G|missense_variant|MODERATE|LOC123790950|LOC123790950|transcript|XM_045795559.1|protein_coding|2/6|c.255C>G|p.Ile85Met|367/1262|255/798|85/265||,A|synonymous_variant|LOW|LOC123
NW_025577149.1  1531554 ANN=A|missense_variant|MODERATE|LOC123790950|LOC123790950|transcript|XM_045795559.1|protein_coding|2/6|c.256G>A|p.Ala86Thr|368/1262|256/798|86/265||,T|missense_variant|MODERATE|LOC
NW_025577149.1  1531556 ANN=C|synonymous_variant|LOW|LOC123790950|LOC123790950|transcript|XM_045795559.1|protein_coding|2/6|c.258T>C|p.Ala86Ala|370/1262|258/798|86/265||
NW_025577149.1  1531560 ANN=G|missense_variant|MODERATE|LOC123790950|LOC123790950|transcript|XM_045795559.1|protein_coding|2/6|c.262T>G|p.Tyr88Asp|374/1262|262/798|88/265||
NW_025577149.1  1531561 ANN=C|missense_variant|MODERATE|LOC123790950|LOC123790950|transcript|XM_045795559.1|protein_coding|2/6|c.263A>C|p.Tyr88Ser|375/1262|263/798|88/265||
NW_025577149.1  1531564 ANN=G|missense_variant|MODERATE|LOC123790950|LOC123790950|transcript|XM_045795559.1|protein_coding|2/6|c.266T>G|p.Leu89Trp|378/1262|266/798|89/265||
NW_025577149.1  1531565 ANN=C|missense_variant|MODERATE|LOC123790950|LOC123790950|transcript|XM_045795559.1|protein_coding|2/6|c.267G>C|p.Leu89Phe|379/1262|267/798|89/265||
NW_025577149.1  1531568 ANN=T|synonymous_variant|LOW|LOC123790950|LOC123790950|transcript|XM_045795559.1|protein_coding|2/6|c.270C>T|p.Asn90Asn|382/1262|270/798|90/265||
NW_025577149.1  1531570 ANN=C|missense_variant|MODERATE|LOC123790950|LOC123790950|transcript|XM_045795559.1|protein_coding|2/6|c.272A>C|p.Gln91Pro|384/1262|272/798|91/265||,G|missense_variant|MODERATE|LOC
NW_025577149.1  1531571 ANN=C|missense_variant|MODERATE|LOC123790950|LOC123790950|transcript|XM_045795559.1|protein_coding|2/6|c.273G>C|p.Gln91His|385/1262|273/798|91/265||,T|missense_variant|MODERATE|LOC
NW_025577149.1  1531574 ANN=T|missense_variant|MODERATE|LOC123790950|LOC123790950|transcript|XM_045795559.1|protein_coding|2/6|c.276G>T|p.Gln92His|388/1262|276/798|92/265||
NW_025577149.1  1531577 ANN=T|missense_variant|MODERATE|LOC123790950|LOC123790950|transcript|XM_045795559.1|protein_coding|2/6|c.279G>T|p.Lys93Asn|391/1262|279/798|93/265||
NW_025577149.1  1531578 ANN=A|missense_variant|MODERATE|LOC123790950|LOC123790950|transcript|XM_045795559.1|protein_coding|2/6|c.280G>A|p.Asp94Asn|392/1262|280/798|94/265||
NW_025577149.1  1531581 ANN=A|missense_variant|MODERATE|LOC123790950|LOC123790950|transcript|XM_045795559.1|protein_coding|2/6|c.283T>A|p.Phe95Ile|395/1262|283/798|95/265||,C|missense_variant|MODERATE|LOC
NW_025577149.1  1531584 ANN=G|missense_variant|MODERATE|LOC123790950|LOC123790950|transcript|XM_045795559.1|protein_coding|2/6|c.286A>G|p.Met96Val|398/1262|286/798|96/265||,T|missense_variant|MODERATE|LOC
NW_025577149.1  1531591 ANN=G|missense_variant|MODERATE|LOC123790950|LOC123790950|transcript|XM_045795559.1|protein_coding|2/6|c.293A>G|p.Gln98Arg|405/1262|293/798|98/265||
NW_025577149.1  1531592 ANN=T|missense_variant|MODERATE|LOC123790950|LOC123790950|transcript|XM_045795559.1|protein_coding|2/6|c.294G>T|p.Gln98His|406/1262|294/798|98/265||
NW_025577149.1  1531594 ANN=A|missense_variant|MODERATE|LOC123790950|LOC123790950|transcript|XM_045795559.1|protein_coding|2/6|c.296C>A|p.Thr99Lys|408/1262|296/798|99/265||,G|missense_variant|MODERATE|LOC
NW_025577149.1  1531596 ANN=A|synonymous_variant|LOW|LOC123790950|LOC123790950|transcript|XM_045795559.1|protein_coding|2/6|c.298C>A|p.Arg100Arg|410/1262|298/798|100/265||
NW_025577149.1  1531597 ANN=T|missense_variant|MODERATE|LOC123790950|LOC123790950|transcript|XM_045795559.1|protein_coding|2/6|c.299G>T|p.Arg100Leu|411/1262|299/798|100/265||
NW_025577149.1  1531599 ANN=T|missense_variant|MODERATE|LOC123790950|LOC123790950|transcript|XM_045795559.1|protein_coding|2/6|c.301G>T|p.Ala101Ser|413/1262|301/798|101/265||
NW_025577149.1  1531603 ANN=A|missense_variant|MODERATE|LOC123790950|LOC123790950|transcript|XM_045795559.1|protein_coding|2/6|c.305C>A|p.Ala102Glu|417/1262|305/798|102/265||
NW_025577149.1  1531608 ANN=T|missense_variant|MODERATE|LOC123790950|LOC123790950|transcript|XM_045795559.1|protein_coding|2/6|c.310G>T|p.Asp104Tyr|422/1262|310/798|104/265||
NW_025577149.1  1531612 ANN=A|missense_variant|MODERATE|LOC123790950|LOC123790950|transcript|XM_045795559.1|protein_coding|2/6|c.314C>A|p.Thr105Lys|426/1262|314/798|105/265||
NW_025577149.1  1531613 ANN=T|synonymous_variant|LOW|LOC123790950|LOC123790950|transcript|XM_045795559.1|protein_coding|2/6|c.315G>T|p.Thr105Thr|427/1262|315/798|105/265||
NW_025577149.1  1531615 ANN=C|missense_variant|MODERATE|LOC123790950|LOC123790950|transcript|XM_045795559.1|protein_coding|2/6|c.317T>C|p.Val106Ala|429/1262|317/798|106/265||
NW_025577149.1  1531628 ANN=G|missense_variant|MODERATE|LOC123790950|LOC123790950|transcript|XM_045795559.1|protein_coding|2/6|c.330C>G|p.Asn110Lys|442/1262|330/798|110/265||
NW_025577149.1  1531632 ANN=G|missense_variant|MODERATE|LOC123790950|LOC123790950|transcript|XM_045795559.1|protein_coding|2/6|c.334C>G|p.Gln112Glu|446/1262|334/798|112/265||
NW_025577149.1  1531635 ANN=T|missense_variant|MODERATE|LOC123790950|LOC123790950|transcript|XM_045795559.1|protein_coding|2/6|c.337A>T|p.Ile113Phe|449/1262|337/798|113/265||
NW_025577149.1  1531636 ANN=C|missense_variant|MODERATE|LOC123790950|LOC123790950|transcript|XM_045795559.1|protein_coding|2/6|c.338T>C|p.Ile113Thr|450/1262|338/798|113/265||
NW_025577149.1  1531637 ANN=A|synonymous_variant|LOW|LOC123790950|LOC123790950|transcript|XM_045795559.1|protein_coding|2/6|c.339T>A|p.Ile113Ile|451/1262|339/798|113/265||
NW_025577149.1  1531638 ANN=T|stop_gained|HIGH|LOC123790950|LOC123790950|transcript|XM_045795559.1|protein_coding|2/6|c.340G>T|p.Glu114*|452/1262|340/798|114/265||;LOF=(LOC123790950|LOC123790950|1|1.00);N
NW_025577149.1  1531645 ANN=T|missense_variant|MODERATE|LOC123790950|LOC123790950|transcript|XM_045795559.1|protein_coding|2/6|c.347G>T|p.Arg116Met|459/1262|347/798|116/265||
NW_025577149.1  1531646 ANN=T|missense_variant|MODERATE|LOC123790950|LOC123790950|transcript|XM_045795559.1|protein_coding|2/6|c.348G>T|p.Arg116Ser|460/1262|348/798|116/265||
NW_025577149.1  1531649 ANN=A|missense_variant|MODERATE|LOC123790950|LOC123790950|transcript|XM_045795559.1|protein_coding|2/6|c.351C>A|p.Phe117Leu|463/1262|351/798|117/265||
NW_025577149.1  1531658 ANN=T|missense_variant|MODERATE|LOC123790950|LOC123790950|transcript|XM_045795559.1|protein_coding|2/6|c.360G>T|p.Gln120His|472/1262|360/798|120/265||
NW_025577149.1  1531659 ANN=T|missense_variant|MODERATE|LOC123790950|LOC123790950|transcript|XM_045795559.1|protein_coding|2/6|c.361C>T|p.Arg121Trp|473/1262|361/798|121/265||
NW_025577149.1  1531666 ANN=A|splice_donor_variant&intron_variant|HIGH|LOC123790950|LOC123790950|transcript|XM_045795559.1|protein_coding|2/5|c.367+1G>A||||||;LOF=(LOC123790950|LOC123790950|1|1.00)
NW_025577149.1  1531678 ANN=T|intron_variant|MODIFIER|LOC123790950|LOC123790950|transcript|XM_045795559.1|protein_coding|2/5|c.367+13G>T||||||
NW_025577149.1  1531679 ANN=C|intron_variant|MODIFIER|LOC123790950|LOC123790950|transcript|XM_045795559.1|protein_coding|2/5|c.367+14T>C||||||
NW_025577149.1  1531686 ANN=A|intron_variant|MODIFIER|LOC123790950|LOC123790950|transcript|XM_045795559.1|protein_coding|2/5|c.367+21C>A||||||
NW_025577149.1  1531687 ANN=A|intron_variant|MODIFIER|LOC123790950|LOC123790950|transcript|XM_045795559.1|protein_coding|2/5|c.367+22C>A||||||
NW_025577149.1  1531688 ANN=A|intron_variant|MODIFIER|LOC123790950|LOC123790950|transcript|XM_045795559.1|protein_coding|2/5|c.367+23C>A||||||,T|intron_variant|MODIFIER|LOC123790950|LOC123790950|transcrip
NW_025577149.1  1531689 ANN=A|intron_variant|MODIFIER|LOC123790950|LOC123790950|transcript|XM_045795559.1|protein_coding|2/5|c.367+24G>A||||||
NW_025577149.1  1531693 ANN=A|intron_variant|MODIFIER|LOC123790950|LOC123790950|transcript|XM_045795559.1|protein_coding|2/5|c.367+28C>A||||||
NW_025577149.1  1531694 ANN=C|intron_variant|MODIFIER|LOC123790950|LOC123790950|transcript|XM_045795559.1|protein_coding|2/5|c.367+29G>C||||||
NW_025577149.1  1531697 ANN=A|intron_variant|MODIFIER|LOC123790950|LOC123790950|transcript|XM_045795559.1|protein_coding|2/5|c.367+32C>A||||||
NW_025577149.1  1531698 ANN=A|intron_variant|MODIFIER|LOC123790950|LOC123790950|transcript|XM_045795559.1|protein_coding|2/5|c.367+33C>A||||||
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The translated portion of exon 2 ranges from 1531403 to 1531659.

Exon 6

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snpEff jax_bbear DQBex6.recode.vcf > DQBex6.ann.vcf
grep -v ^# DQBex6.ann.vcf | cut -f 1,2,8
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NW_025577149.1  1535166 ANN=G|splice_region_variant&intron_variant|LOW|LOC123790950|LOC123790950|transcript|XM_045795559.1|protein_coding|5/5|c.785-4A>G||||||
NW_025577149.1  1535171 ANN=A|splice_region_variant&synonymous_variant|LOW|LOC123790950|LOC123790950|transcript|XM_045795559.1|protein_coding|6/6|c.786G>A|p.Gly262Gly|898/1262|786/798|262/265||
NW_025577149.1  1535177 ANN=A|synonymous_variant|LOW|LOC123790950|LOC123790950|transcript|XM_045795559.1|protein_coding|6/6|c.792G>A|p.Leu264Leu|904/1262|792/798|264/265||
NW_025577149.1  1535179 ANN=A|missense_variant|MODERATE|LOC123790950|LOC123790950|transcript|XM_045795559.1|protein_coding|6/6|c.794G>A|p.Arg265His|906/1262|794/798|265/265||,T|missense_variant|MODERATE|L
NW_025577149.1  1535184 ANN=T|3_prime_UTR_variant|MODIFIER|LOC123790950|LOC123790950|transcript|XM_045795559.1|protein_coding|6/6|c.*1C>T|||||1|
NW_025577149.1  1535185 ANN=T|3_prime_UTR_variant|MODIFIER|LOC123790950|LOC123790950|transcript|XM_045795559.1|protein_coding|6/6|c.*2G>T|||||2|
NW_025577149.1  1535187 ANN=A|3_prime_UTR_variant|MODIFIER|LOC123790950|LOC123790950|transcript|XM_045795559.1|protein_coding|6/6|c.*4C>A|||||4|
NW_025577149.1  1535188 ANN=C|3_prime_UTR_variant|MODIFIER|LOC123790950|LOC123790950|transcript|XM_045795559.1|protein_coding|6/6|c.*5T>C|||||5|
NW_025577149.1  1535196 ANN=T|3_prime_UTR_variant|MODIFIER|LOC123790950|LOC123790950|transcript|XM_045795559.1|protein_coding|6/6|c.*13C>T|||||13|
NW_025577149.1  1535198 ANN=C|3_prime_UTR_variant|MODIFIER|LOC123790950|LOC123790950|transcript|XM_045795559.1|protein_coding|6/6|c.*15T>C|||||15|
NW_025577149.1  1535201 ANN=T|3_prime_UTR_variant|MODIFIER|LOC123790950|LOC123790950|transcript|XM_045795559.1|protein_coding|6/6|c.*18G>T|||||18|
NW_025577149.1  1535205 ANN=A|3_prime_UTR_variant|MODIFIER|LOC123790950|LOC123790950|transcript|XM_045795559.1|protein_coding|6/6|c.*22C>A|||||22|,T|3_prime_UTR_variant|MODIFIER|LOC123790950|LOC123790950|
NW_025577149.1  1535229 ANN=A|3_prime_UTR_variant|MODIFIER|LOC123790950|LOC123790950|transcript|XM_045795559.1|protein_coding|6/6|c.*46C>A|||||46|
NW_025577149.1  1535230 ANN=A|3_prime_UTR_variant|MODIFIER|LOC123790950|LOC123790950|transcript|XM_045795559.1|protein_coding|6/6|c.*47C>A|||||47|
NW_025577149.1  1535233 ANN=T|3_prime_UTR_variant|MODIFIER|LOC123790950|LOC123790950|transcript|XM_045795559.1|protein_coding|6/6|c.*50G>T|||||50|
NW_025577149.1  1535236 ANN=A|3_prime_UTR_variant|MODIFIER|LOC123790950|LOC123790950|transcript|XM_045795559.1|protein_coding|6/6|c.*53C>A|||||53|
NW_025577149.1  1535250 ANN=A|3_prime_UTR_variant|MODIFIER|LOC123790950|LOC123790950|transcript|XM_045795559.1|protein_coding|6/6|c.*67G>A|||||67|
NW_025577149.1  1535257 ANN=C|3_prime_UTR_variant|MODIFIER|LOC123790950|LOC123790950|transcript|XM_045795559.1|protein_coding|6/6|c.*74T>C|||||74|
NW_025577149.1  1535267 ANN=A|3_prime_UTR_variant|MODIFIER|LOC123790950|LOC123790950|transcript|XM_045795559.1|protein_coding|6/6|c.*84C>A|||||84|
NW_025577149.1  1535268 ANN=A|3_prime_UTR_variant|MODIFIER|LOC123790950|LOC123790950|transcript|XM_045795559.1|protein_coding|6/6|c.*85C>A|||||85|
NW_025577149.1  1535275 ANN=G|3_prime_UTR_variant|MODIFIER|LOC123790950|LOC123790950|transcript|XM_045795559.1|protein_coding|6/6|c.*92A>G|||||92|
NW_025577149.1  1535298 ANN=C|3_prime_UTR_variant|MODIFIER|LOC123790950|LOC123790950|transcript|XM_045795559.1|protein_coding|6/6|c.*115G>C|||||115|
NW_025577149.1  1535306 ANN=T|3_prime_UTR_variant|MODIFIER|LOC123790950|LOC123790950|transcript|XM_045795559.1|protein_coding|6/6|c.*123G>T|||||123|
NW_025577149.1  1535309 ANN=A|3_prime_UTR_variant|MODIFIER|LOC123790950|LOC123790950|transcript|XM_045795559.1|protein_coding|6/6|c.*126C>A|||||126|
NW_025577149.1  1535310 ANN=A|3_prime_UTR_variant|MODIFIER|LOC123790950|LOC123790950|transcript|XM_045795559.1|protein_coding|6/6|c.*127G>A|||||127|,T|3_prime_UTR_variant|MODIFIER|LOC123790950|LOC12379095
NW_025577149.1  1535311 ANN=A|3_prime_UTR_variant|MODIFIER|LOC123790950|LOC123790950|transcript|XM_045795559.1|protein_coding|6/6|c.*128G>A|||||128|
NW_025577149.1  1535318 ANN=T|3_prime_UTR_variant|MODIFIER|LOC123790950|LOC123790950|transcript|XM_045795559.1|protein_coding|6/6|c.*135C>T|||||135|
NW_025577149.1  1535321 ANN=A|3_prime_UTR_variant|MODIFIER|LOC123790950|LOC123790950|transcript|XM_045795559.1|protein_coding|6/6|c.*138G>A|||||138|
```

```
NW_025577149.1  1535325 ANN=G|3_prime_UTR_variant|MODIFIER|LOC123790950|LOC123790950|transcript|XM_045795559.1|protein_coding|6/6|c.*142C>G|||||142|
NW_025577149.1  1535326 ANN=T|3_prime_UTR_variant|MODIFIER|LOC123790950|LOC123790950|transcript|XM_045795559.1|protein_coding|6/6|c.*143C>T|||||143|
NW_025577149.1  1535327 ANN=A|3_prime_UTR_variant|MODIFIER|LOC123790950|LOC123790950|transcript|XM_045795559.1|protein_coding|6/6|c.*144C>A|||||144|
NW_025577149.1  1535330 ANN=A|3_prime_UTR_variant|MODIFIER|LOC123790950|LOC123790950|transcript|XM_045795559.1|protein_coding|6/6|c.*147G>A|||||147|
NW_025577149.1  1535332 ANN=A|3_prime_UTR_variant|MODIFIER|LOC123790950|LOC123790950|transcript|XM_045795559.1|protein_coding|6/6|c.*149C>A|||||149|,T|3_prime_UTR_variant|MODIFIER|LOC123790950|LOC12379095
NW_025577149.1  1535333 ANN=A|3_prime_UTR_variant|MODIFIER|LOC123790950|LOC123790950|transcript|XM_045795559.1|protein_coding|6/6|c.*150C>A|||||150|
NW_025577149.1  1535335 ANN=A|3_prime_UTR_variant|MODIFIER|LOC123790950|LOC123790950|transcript|XM_045795559.1|protein_coding|6/6|c.*152G>A|||||152|
NW_025577149.1  1535344 ANN=T|3_prime_UTR_variant|MODIFIER|LOC123790950|LOC123790950|transcript|XM_045795559.1|protein_coding|6/6|c.*161G>T|||||161|
NW_025577149.1  1535351 ANN=A|3_prime_UTR_variant|MODIFIER|LOC123790950|LOC123790950|transcript|XM_045795559.1|protein_coding|6/6|c.*168T>A|||||168|
NW_025577149.1  1535352 ANN=T|3_prime_UTR_variant|MODIFIER|LOC123790950|LOC123790950|transcript|XM_045795559.1|protein_coding|6/6|c.*169G>T|||||169|
NW_025577149.1  1535355 ANN=T|3_prime_UTR_variant|MODIFIER|LOC123790950|LOC123790950|transcript|XM_045795559.1|protein_coding|6/6|c.*172G>T|||||172|
NW_025577149.1  1535363 ANN=G|3_prime_UTR_variant|MODIFIER|LOC123790950|LOC123790950|transcript|XM_045795559.1|protein_coding|6/6|c.*180A>G|||||180|
NW_025577149.1  1535364 ANN=A|3_prime_UTR_variant|MODIFIER|LOC123790950|LOC123790950|transcript|XM_045795559.1|protein_coding|6/6|c.*181C>A|||||181|
NW_025577149.1  1535378 ANN=C|3_prime_UTR_variant|MODIFIER|LOC123790950|LOC123790950|transcript|XM_045795559.1|protein_coding|6/6|c.*195T>C|||||195|
NW_025577149.1  1535382 ANN=T|3_prime_UTR_variant|MODIFIER|LOC123790950|LOC123790950|transcript|XM_045795559.1|protein_coding|6/6|c.*199C>T|||||199|
NW_025577149.1  1535388 ANN=T|3_prime_UTR_variant|MODIFIER|LOC123790950|LOC123790950|transcript|XM_045795559.1|protein_coding|6/6|c.*205C>T|||||205|
NW_025577149.1  1535390 ANN=A|3_prime_UTR_variant|MODIFIER|LOC123790950|LOC123790950|transcript|XM_045795559.1|protein_coding|6/6|c.*207C>A|||||207|
NW_025577149.1  1535392 ANN=A|3_prime_UTR_variant|MODIFIER|LOC123790950|LOC123790950|transcript|XM_045795559.1|protein_coding|6/6|c.*209C>A|||||209|
NW_025577149.1  1535394 ANN=A|3_prime_UTR_variant|MODIFIER|LOC123790950|LOC123790950|transcript|XM_045795559.1|protein_coding|6/6|c.*211C>A|||||211|
NW_025577149.1  1535396 ANN=T|3_prime_UTR_variant|MODIFIER|LOC123790950|LOC123790950|transcript|XM_045795559.1|protein_coding|6/6|c.*213G>T|||||213|
NW_025577149.1  1535397 ANN=T|3_prime_UTR_variant|MODIFIER|LOC123790950|LOC123790950|transcript|XM_045795559.1|protein_coding|6/6|c.*214C>T|||||214|
NW_025577149.1  1535407 ANN=C|3_prime_UTR_variant|MODIFIER|LOC123790950|LOC123790950|transcript|XM_045795559.1|protein_coding|6/6|c.*224G>C|||||224|
NW_025577149.1  1535414 ANN=G|3_prime_UTR_variant|MODIFIER|LOC123790950|LOC123790950|transcript|XM_045795559.1|protein_coding|6/6|c.*231A>G|||||231|
NW_025577149.1  1535418 ANN=A|3_prime_UTR_variant|MODIFIER|LOC123790950|LOC123790950|transcript|XM_045795559.1|protein_coding|6/6|c.*235C>A|||||235|
NW_025577149.1  1535423 ANN=T|3_prime_UTR_variant|MODIFIER|LOC123790950|LOC123790950|transcript|XM_045795559.1|protein_coding|6/6|c.*240G>T|||||240|
NW_025577149.1  1535432 ANN=T|3_prime_UTR_variant|MODIFIER|LOC123790950|LOC123790950|transcript|XM_045795559.1|protein_coding|6/6|c.*249G>T|||||249|
NW_025577149.1  1535441 ANN=A|3_prime_UTR_variant|MODIFIER|LOC123790950|LOC123790950|transcript|XM_045795559.1|protein_coding|6/6|c.*258C>A|||||258|
NW_025577149.1  1535444 ANN=C|3_prime_UTR_variant|MODIFIER|LOC123790950|LOC123790950|transcript|XM_045795559.1|protein_coding|6/6|c.*261T>C|||||261|
NW_025577149.1  1535447 ANN=G|3_prime_UTR_variant|MODIFIER|LOC123790950|LOC123790950|transcript|XM_045795559.1|protein_coding|6/6|c.*264C>G|||||264|
NW_025577149.1  1535448 ANN=A|3_prime_UTR_variant|MODIFIER|LOC123790950|LOC123790950|transcript|XM_045795559.1|protein_coding|6/6|c.*265C>A|||||265|
NW_025577149.1  1535450 ANN=A|3_prime_UTR_variant|MODIFIER|LOC123790950|LOC123790950|transcript|XM_045795559.1|protein_coding|6/6|c.*267C>A|||||267|
NW_025577149.1  1535451 ANN=C|3_prime_UTR_variant|MODIFIER|LOC123790950|LOC123790950|transcript|XM_045795559.1|protein_coding|6/6|c.*268A>C|||||268|,T|3_prime_UTR_variant|MODIFIER|LOC123790950|LOC12379095
NW_025577149.1  1535452 ANN=T|3_prime_UTR_variant|MODIFIER|LOC123790950|LOC123790950|transcript|XM_045795559.1|protein_coding|6/6|c.*269G>T|||||269|
NW_025577149.1  1535459 ANN=A|3_prime_UTR_variant|MODIFIER|LOC123790950|LOC123790950|transcript|XM_045795559.1|protein_coding|6/6|c.*276C>A|||||276|
NW_025577149.1  1535464 ANN=T|3_prime_UTR_variant|MODIFIER|LOC123790950|LOC123790950|transcript|XM_045795559.1|protein_coding|6/6|c.*281G>T|||||281|
NW_025577149.1  1535465 ANN=T|3_prime_UTR_variant|MODIFIER|LOC123790950|LOC123790950|transcript|XM_045795559.1|protein_coding|6/6|c.*282G>T|||||282|
NW_025577149.1  1535474 ANN=A|3_prime_UTR_variant|MODIFIER|LOC123790950|LOC123790950|transcript|XM_045795559.1|protein_coding|6/6|c.*291G>A|||||291|
NW_025577149.1  1535476 ANN=G|3_prime_UTR_variant|MODIFIER|LOC123790950|LOC123790950|transcript|XM_045795559.1|protein_coding|6/6|c.*293A>G|||||293|
NW_025577149.1  1535481 ANN=A|3_prime_UTR_variant|MODIFIER|LOC123790950|LOC123790950|transcript|XM_045795559.1|protein_coding|6/6|c.*298T>A|||||298|
NW_025577149.1  1535486 ANN=C|3_prime_UTR_variant|MODIFIER|LOC123790950|LOC123790950|transcript|XM_045795559.1|protein_coding|6/6|c.*303T>C|||||303|
NW_025577149.1  1535488 ANN=T|3_prime_UTR_variant|MODIFIER|LOC123790950|LOC123790950|transcript|XM_045795559.1|protein_coding|6/6|c.*305G>T|||||305|
NW_025577149.1  1535495 ANN=A|3_prime_UTR_variant|MODIFIER|LOC123790950|LOC123790950|transcript|XM_045795559.1|protein_coding|6/6|c.*312G>A|||||312|
NW_025577149.1  1535499 ANN=A|3_prime_UTR_variant|MODIFIER|LOC123790950|LOC123790950|transcript|XM_045795559.1|protein_coding|6/6|c.*316T>A|||||316|
NW_025577149.1  1535505 ANN=A|3_prime_UTR_variant|MODIFIER|LOC123790950|LOC123790950|transcript|XM_045795559.1|protein_coding|6/6|c.*322C>A|||||322|
NW_025577149.1  1535513 ANN=C|3_prime_UTR_variant|MODIFIER|LOC123790950|LOC123790950|transcript|XM_045795559.1|protein_coding|6/6|c.*330A>C|||||330|
NW_025577149.1  1535522 ANN=A|3_prime_UTR_variant|MODIFIER|LOC123790950|LOC123790950|transcript|XM_045795559.1|protein_coding|6/6|c.*339T>A|||||339|
NW_025577149.1  1535529 ANN=A|3_prime_UTR_variant|MODIFIER|LOC123790950|LOC123790950|transcript|XM_045795559.1|protein_coding|6/6|c.*346C>A|||||346|
NW_025577149.1  1535531 ANN=A|3_prime_UTR_variant|MODIFIER|LOC123790950|LOC123790950|transcript|XM_045795559.1|protein_coding|6/6|c.*348C>A|||||348|
```

Most of the exon 6 chunk in our dataset is in a 3' untranslated region (aside from 1535177 to 1535179).

Extract *only* the exons

Note that we lose the annotation information when we subset to these regions, so we'll have to annotate the SNPs retained after filtering the `.vcf` again later. Our strategy will be to i) identify translated portions of the exons (done above), ii) trim to those regions, iii) apply filters for quality, genotyping depth, etc., iv) phase the filtered SNPs, and v) re-annotate the loci that pass our filters.

Exon 2

```
echo 'CHROM START END' > DQBex2Only.bed
echo 'NW_025577149.1 1531402 1531659' >> DQBex2Only.bed
vcftools --vcf DQBex2.recode.vcf --bed DQBex2Only.bed \
         --out DQBex2.sub --recode
```

```
VCFtools - 0.1.17
(C) Adam Auton and Anthony Marcketta 2009

Parameters as interpreted:
    --vcf DQBex2.recode.vcf
    --out DQBex2.sub
    --recode
    --bed DQBex2Only.bed
```

```
Warning: Expected at least 2 parts in INFO entry: ID=AF,Number=A,Type=Float,Description="Estimated allel
Warning: Expected at least 2 parts in INFO entry: ID=PRO,Number=1,Type=Float,Description="Reference all
Warning: Expected at least 2 parts in INFO entry: ID=PAO,Number=A,Type=Float,Description="Alternate all
Warning: Expected at least 2 parts in INFO entry: ID=SRP,Number=1,Type=Float,Description="Strand balanc
Warning: Expected at least 2 parts in INFO entry: ID=SAP,Number=A,Type=Float,Description="Strand balanc
Warning: Expected at least 2 parts in INFO entry: ID=AB,Number=A,Type=Float,Description="Allele balance
Warning: Expected at least 2 parts in INFO entry: ID=ABP,Number=A,Type=Float,Description="Allele balanc
Warning: Expected at least 2 parts in INFO entry: ID=RPP,Number=A,Type=Float,Description="Read Placemen
Warning: Expected at least 2 parts in INFO entry: ID=RPPR,Number=1,Type=Float,Description="Read Placeme
Warning: Expected at least 2 parts in INFO entry: ID=EPP,Number=A,Type=Float,Description="End Placement
Warning: Expected at least 2 parts in INFO entry: ID=EPPR,Number=1,Type=Float,Description="End Placemen
Warning: Expected at least 2 parts in INFO entry: ID=TYPE,Number=A,Type=String,Description="The type of
Warning: Expected at least 2 parts in INFO entry: ID=TYPE,Number=A,Type=String,Description="The type of
Warning: Expected at least 2 parts in INFO entry: ID=TYPE,Number=A,Type=String,Description="The type of
Warning: Expected at least 2 parts in INFO entry: ID=TYPE,Number=A,Type=String,Description="The type of
Warning: Expected at least 2 parts in INFO entry: ID=TYPE,Number=A,Type=String,Description="The type of
Warning: Expected at least 2 parts in INFO entry: ID=CIGAR,Number=A,Type=String,Description="The extend
Warning: Expected at least 2 parts in FORMAT entry: ID=GQ,Number=1,Type=Float,Description="Genotype Qual
Warning: Expected at least 2 parts in FORMAT entry: ID=GL,Number=G,Type=Float,Description="Genotype Lik
After filtering, kept 260 out of 260 Individuals
Outputting VCF file...
    Read 2 BED file entries.
After filtering, kept 105 out of a possible 144 Sites
Run Time = 0.00 seconds
```

Exon 6

```
echo 'CHROM START END' > DQBex6Only.bed
echo 'NW_025577149.1 1535176 1535179' >> DQBex6Only.bed
vcftools --vcf DQBex6.recode.vcf --bed DQBex6Only.bed \
          --out DQBex6.sub --recode
```

```
VCFtools - 0.1.17
(C) Adam Auton and Anthony Marcketta 2009

Parameters as interpreted:
    --vcf DQBex6.recode.vcf
    --out DQBex6.sub
    --recode
    --bed DQBex6Only.bed

Warning: Expected at least 2 parts in INFO entry: ID=AF,Number=A,Type=Float,Description="Estimated allel
Warning: Expected at least 2 parts in INFO entry: ID=PRO,Number=1,Type=Float,Description="Reference all
Warning: Expected at least 2 parts in INFO entry: ID=PAO,Number=A,Type=Float,Description="Alternate all
Warning: Expected at least 2 parts in INFO entry: ID=SRP,Number=1,Type=Float,Description="Strand balanc
Warning: Expected at least 2 parts in INFO entry: ID=SAP,Number=A,Type=Float,Description="Strand balanc
Warning: Expected at least 2 parts in INFO entry: ID=AB,Number=A,Type=Float,Description="Allele balance
Warning: Expected at least 2 parts in INFO entry: ID=ABP,Number=A,Type=Float,Description="Allele balanc
Warning: Expected at least 2 parts in INFO entry: ID=RPP,Number=A,Type=Float,Description="Read Placemen
Warning: Expected at least 2 parts in INFO entry: ID=RPPR,Number=1,Type=Float,Description="Read Placeme
```

```
Warning: Expected at least 2 parts in INFO entry: ID=EPP,Number=A,Type=Float,Description="End Placement
Warning: Expected at least 2 parts in INFO entry: ID=EPPR,Number=1,Type=Float,Description="End Placemen
Warning: Expected at least 2 parts in INFO entry: ID=TYPE,Number=A,Type=String,Description="The type of
Warning: Expected at least 2 parts in INFO entry: ID=TYPE,Number=A,Type=String,Description="The type of
Warning: Expected at least 2 parts in INFO entry: ID=TYPE,Number=A,Type=String,Description="The type of
Warning: Expected at least 2 parts in INFO entry: ID=TYPE,Number=A,Type=String,Description="The type of
Warning: Expected at least 2 parts in INFO entry: ID=TYPE,Number=A,Type=String,Description="The type of
Warning: Expected at least 2 parts in INFO entry: ID=CIGAR,Number=A,Type=String,Description="The extend
Warning: Expected at least 2 parts in FORMAT entry: ID=GQ,Number=1,Type=Float,Description="Genotype Qua
Warning: Expected at least 2 parts in FORMAT entry: ID=GL,Number=G,Type=Float,Description="Genotype Lik
After filtering, kept 260 out of 260 Individuals
Outputting VCF file...
    Read 2 BED file entries.
After filtering, kept 2 out of a possible 76 Sites
Run Time = 1.00 seconds
```

Filter the dataset

Note that *missing data are imputed* during the statistical phasing in `beagle`. With that in mind, we'll impose a fairly restrictive filter on missing data. This will minimize the need to impute genotypes.

Filtering *DQB* Exon 2

Now filter for genotype quality ($Q \geq 30$), genotype depth ($DP \geq 10$ reads), and minor allele count (mac ≥ 2. The last of these filters will remove singleton variants, so we will underestimate haplotype diversity in the analyses below. However, the accuracy of statistical phasing is low for singleton SNPs, so the information lost is not too great. Furthermore, we are removing singletons across the entire dataset (rather than population-by-population).

```
#Exon 2
#Site Quality, Genotyping Depth, Missingness, Minor Allele Count
vcftools --vcf DQBex2.sub.recode.vcf --out DQBex2_filt1 \
          --minQ 30 --minDP 10 --mac 2 \
          --remove-filtered-all --recode

#Check frequencies
vcftools --vcf DQBex2_filt1.recode.vcf --out DQBex2_filt1 \
          --freq
```

```
VCFtools - 0.1.17
(C) Adam Auton and Anthony Marcketta 2009

Parameters as interpreted:
    --vcf DQBex2.sub.recode.vcf
    --mac 2
    --minDP 10
    --minQ 30
    --out DQBex2_filt1
    --recode
    --remove-filtered-all
```

```
Warning: Expected at least 2 parts in INFO entry: ID=AF,Number=A,Type=Float,Description="Estimated allel
Warning: Expected at least 2 parts in INFO entry: ID=PRO,Number=1,Type=Float,Description="Reference allo
Warning: Expected at least 2 parts in INFO entry: ID=PAO,Number=A,Type=Float,Description="Alternate allo
Warning: Expected at least 2 parts in INFO entry: ID=SRP,Number=1,Type=Float,Description="Strand balance
Warning: Expected at least 2 parts in INFO entry: ID=SAP,Number=A,Type=Float,Description="Strand balance
Warning: Expected at least 2 parts in INFO entry: ID=AB,Number=A,Type=Float,Description="Allele balance
Warning: Expected at least 2 parts in INFO entry: ID=ABP,Number=A,Type=Float,Description="Allele balance
Warning: Expected at least 2 parts in INFO entry: ID=RPP,Number=A,Type=Float,Description="Read Placement
Warning: Expected at least 2 parts in INFO entry: ID=RPPR,Number=1,Type=Float,Description="Read Placemer
Warning: Expected at least 2 parts in INFO entry: ID=EPP,Number=A,Type=Float,Description="End Placement
Warning: Expected at least 2 parts in INFO entry: ID=EPPR,Number=1,Type=Float,Description="End Placement
Warning: Expected at least 2 parts in INFO entry: ID=TYPE,Number=A,Type=String,Description="The type of
Warning: Expected at least 2 parts in INFO entry: ID=TYPE,Number=A,Type=String,Description="The type of
Warning: Expected at least 2 parts in INFO entry: ID=TYPE,Number=A,Type=String,Description="The type of
Warning: Expected at least 2 parts in INFO entry: ID=TYPE,Number=A,Type=String,Description="The type of
Warning: Expected at least 2 parts in INFO entry: ID=TYPE,Number=A,Type=String,Description="The type of
Warning: Expected at least 2 parts in INFO entry: ID=CIGAR,Number=A,Type=String,Description="The extende
Warning: Expected at least 2 parts in FORMAT entry: ID=GQ,Number=1,Type=Float,Description="Genotype Qual
Warning: Expected at least 2 parts in FORMAT entry: ID=GL,Number=G,Type=Float,Description="Genotype Like
After filtering, kept 260 out of 260 Individuals
Outputting VCF file...
After filtering, kept 38 out of a possible 105 Sites
Run Time = 0.00 seconds

VCFtools - 0.1.17
(C) Adam Auton and Anthony Marcketta 2009

Parameters as interpreted:
    --vcf DQBex2_filt1.recode.vcf
    --freq
    --out DQBex2_filt1

Warning: Expected at least 2 parts in INFO entry: ID=AF,Number=A,Type=Float,Description="Estimated allel
Warning: Expected at least 2 parts in INFO entry: ID=PRO,Number=1,Type=Float,Description="Reference allo
Warning: Expected at least 2 parts in INFO entry: ID=PAO,Number=A,Type=Float,Description="Alternate allo
Warning: Expected at least 2 parts in INFO entry: ID=SRP,Number=1,Type=Float,Description="Strand balance
Warning: Expected at least 2 parts in INFO entry: ID=SAP,Number=A,Type=Float,Description="Strand balance
Warning: Expected at least 2 parts in INFO entry: ID=AB,Number=A,Type=Float,Description="Allele balance
Warning: Expected at least 2 parts in INFO entry: ID=ABP,Number=A,Type=Float,Description="Allele balance
Warning: Expected at least 2 parts in INFO entry: ID=RPP,Number=A,Type=Float,Description="Read Placement
Warning: Expected at least 2 parts in INFO entry: ID=RPPR,Number=1,Type=Float,Description="Read Placemer
Warning: Expected at least 2 parts in INFO entry: ID=EPP,Number=A,Type=Float,Description="End Placement
Warning: Expected at least 2 parts in INFO entry: ID=EPPR,Number=1,Type=Float,Description="End Placement
Warning: Expected at least 2 parts in INFO entry: ID=TYPE,Number=A,Type=String,Description="The type of
Warning: Expected at least 2 parts in INFO entry: ID=TYPE,Number=A,Type=String,Description="The type of
Warning: Expected at least 2 parts in INFO entry: ID=TYPE,Number=A,Type=String,Description="The type of
Warning: Expected at least 2 parts in INFO entry: ID=TYPE,Number=A,Type=String,Description="The type of
Warning: Expected at least 2 parts in INFO entry: ID=TYPE,Number=A,Type=String,Description="The type of
Warning: Expected at least 2 parts in INFO entry: ID=CIGAR,Number=A,Type=String,Description="The extende
Warning: Expected at least 2 parts in FORMAT entry: ID=GQ,Number=1,Type=Float,Description="Genotype Qual
Warning: Expected at least 2 parts in FORMAT entry: ID=GL,Number=G,Type=Float,Description="Genotype Like
After filtering, kept 260 out of 260 Individuals
Outputting Frequency Statistics...
After filtering, kept 38 out of a possible 38 Sites
```

```
Run Time = 0.00 seconds
```

Now ID bi-allelic sites from the frequencies file. . .

```r
#Retain only bi-allelic loci
snpfreq <- read.table("DQBex2_filt1.frq", fill = TRUE)
snpfreq <- snpfreq[-1,]
colnames(snpfreq) <- c("CHROM","POS","N_ALLELES","N_CHR", "A1","A2","A3")
snpfreq
```

```
##              CHROM     POS N_ALLELES N_CHR         A1           A2           A3
## 2  NW_025577149.1 1531404         2   450   G:0.995556 T:0.00444444
## 3  NW_025577149.1 1531407         3   452   C:0.986726 G:0.00663717 T:0.00663717
## 4  NW_025577149.1 1531422         3   348   G:0.985632 A:0.00574713 C:0.00862069
## 5  NW_025577149.1 1531423         2   348   A:0.991379 T:0.00862069
## 6  NW_025577149.1 1531451         2   464   G:0.987069   C:0.012931
## 7  NW_025577149.1 1531454         2   464   G:0.778017   A:0.221983
## 8  NW_025577149.1 1531458         2   460   G:0.271739   C:0.728261
## 9  NW_025577149.1 1531459         2   460   G:0.271739   T:0.728261
## 10 NW_025577149.1 1531464         2   460   A:0.758696   G:0.241304
## 11 NW_025577149.1 1531465         2   458   C:0.770742   A:0.229258
## 12 NW_025577149.1 1531470         2   458 C:0.0305677   T:0.969432
## 13 NW_025577149.1 1531471         2   458   A:0.995633 C:0.00436681
## 14 NW_025577149.1 1531492         2   478   T:0.439331   A:0.560669
## 15 NW_025577149.1 1531495         3   480   T:0.572917    C:0.41875 G:0.00833333
## 16 NW_025577149.1 1531501         2   472   T:0.959746  A:0.0402542
## 17 NW_025577149.1 1531506         2   470   A:0.968085  G:0.0319149
## 18 NW_025577149.1 1531507         2   470   G:0.965957  A:0.0340426
## 19 NW_025577149.1 1531514         2   464   G:0.991379 T:0.00862069
## 20 NW_025577149.1 1531521         2   458 C:0.0393013   T:0.960699
## 21 NW_025577149.1 1531525         2   108   G:0.953704  A:0.0462963
## 22 NW_025577149.1 1531527         2   346   G:0.973988  C:0.0260116
## 23 NW_025577149.1 1531535         2   444   G:0.984234  C:0.0157658
## 24 NW_025577149.1 1531550         2   384   C:0.802083   G:0.197917
## 25 NW_025577149.1 1531552         3    86   T:0.790698   A:0.127907   C:0.0813953
## 26 NW_025577149.1 1531556         2   318   T:0.987421  C:0.0125786
## 27 NW_025577149.1 1531560         2   418   T:0.397129   G:0.602871
## 28 NW_025577149.1 1531561         2   426   A:0.410798   C:0.589202
## 29 NW_025577149.1 1531564         2   422   T:0.125592   G:0.874408
## 30 NW_025577149.1 1531565         2   430   G:0.890698   C:0.109302
## 31 NW_025577149.1 1531570         3   426   A:0.699531    C:0.29108 G:0.00938967
## 32 NW_025577149.1 1531574         2   442    G:0.99095 T:0.00904977
## 33 NW_025577149.1 1531581         3   442 T:0.0633484   C:0.477376    A:0.459276
## 34 NW_025577149.1 1531584         3   444   A:0.515766   G:0.463964   T:0.0202703
## 35 NW_025577149.1 1531591         2   448 A:0.0892857   G:0.910714
## 36 NW_025577149.1 1531594         3   454   C:0.515419   G:0.455947   A:0.0286344
## 37 NW_025577149.1 1531603         2   454 C:0.0704846   A:0.929515
## 38 NW_025577149.1 1531612         2   460   C:0.986957  A:0.0130435
## 39 NW_025577149.1 1531646         2   440   G:0.636364   T:0.363636
```

```r
biallelic <- snpfreq[which(snpfreq$N_ALLELES == 2),]
nrow(biallelic)
```

```
## [1] 30
```

```r
fixed.sites <- c(grep(":1", biallelic$A1),
                 grep(":1", biallelic$A2),
                 grep(":1", biallelic$A3))

if(length(fixed.sites) > 0) {
  fixed.sites
  biallelic[fixed.sites,]
  confirmed.biallelic <- biallelic[-fixed.sites,c(1,2)]
} else {
  confirmed.biallelic <- biallelic
}

nrow(confirmed.biallelic)
```

```
## [1] 30
```

```r
write.table(confirmed.biallelic, file = "DQBex2_biallelic.txt",
            quote = FALSE, row.names = FALSE, col.names = FALSE)
```

And filter to keep only those sites...

```bash
#Biallelic SNPs
vcftools --vcf DQBex2_filt1.recode.vcf --out DQBex2_filt2 \
         --positions DQBex2_biallelic.txt \
         --remove-filtered-all --recode

#Check missing data levels
vcftools --vcf DQBex2_filt2.recode.vcf --out DQBex2_filt2 \
         --missing-indv
```

```
VCFtools - 0.1.17
(C) Adam Auton and Anthony Marcketta 2009

Parameters as interpreted:
    --vcf DQBex2_filt1.recode.vcf
    --out DQBex2_filt2
    --positions DQBex2_biallelic.txt
    --recode
    --remove-filtered-all

Warning: Expected at least 2 parts in INFO entry: ID=AF,Number=A,Type=Float,Description="Estimated alle
Warning: Expected at least 2 parts in INFO entry: ID=PRO,Number=1,Type=Float,Description="Reference all
Warning: Expected at least 2 parts in INFO entry: ID=PAO,Number=A,Type=Float,Description="Alternate all
Warning: Expected at least 2 parts in INFO entry: ID=SRP,Number=1,Type=Float,Description="Strand balanc
Warning: Expected at least 2 parts in INFO entry: ID=SAP,Number=A,Type=Float,Description="Strand balanc
Warning: Expected at least 2 parts in INFO entry: ID=AB,Number=A,Type=Float,Description="Allele balance
Warning: Expected at least 2 parts in INFO entry: ID=ABP,Number=A,Type=Float,Description="Allele balanc
Warning: Expected at least 2 parts in INFO entry: ID=RPP,Number=A,Type=Float,Description="Read Placemen
Warning: Expected at least 2 parts in INFO entry: ID=RPPR,Number=1,Type=Float,Description="Read Placeme
Warning: Expected at least 2 parts in INFO entry: ID=EPP,Number=A,Type=Float,Description="End Placement
Warning: Expected at least 2 parts in INFO entry: ID=EPPR,Number=1,Type=Float,Description="End Placemen
```

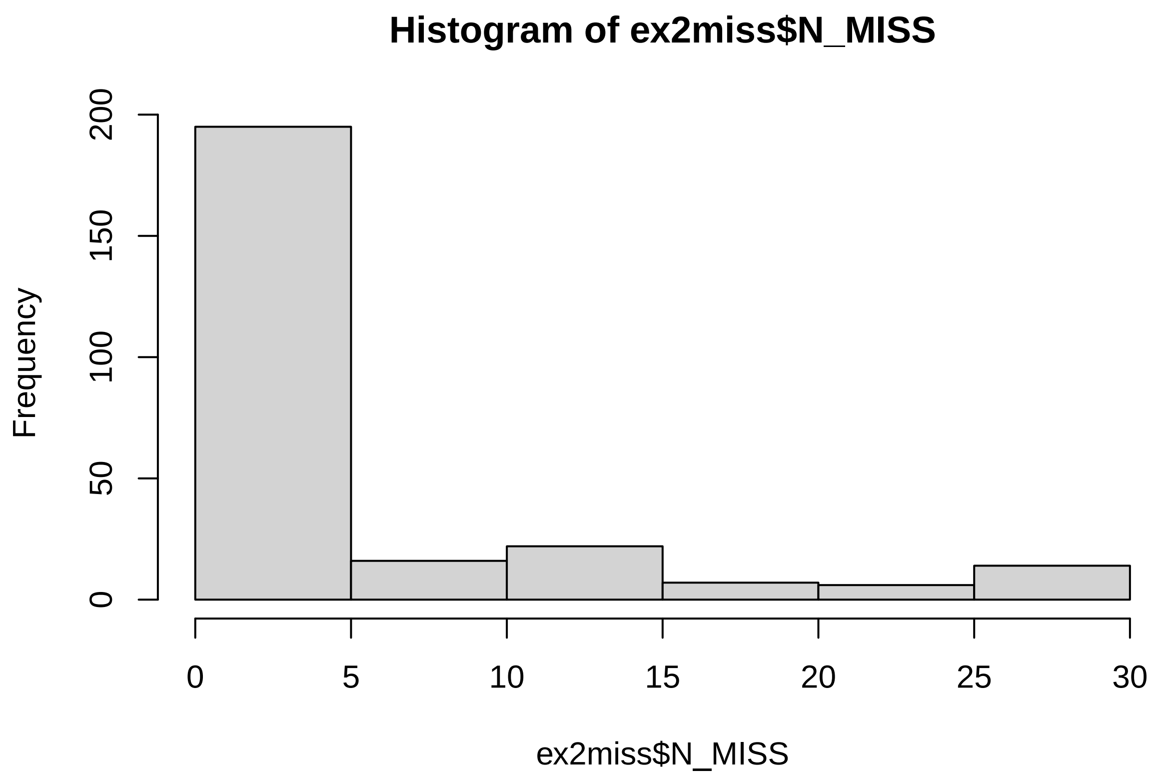
```
Warning: Expected at least 2 parts in INFO entry: ID=TYPE,Number=A,Type=String,Description="The type of
Warning: Expected at least 2 parts in INFO entry: ID=TYPE,Number=A,Type=String,Description="The type of
Warning: Expected at least 2 parts in INFO entry: ID=TYPE,Number=A,Type=String,Description="The type of
Warning: Expected at least 2 parts in INFO entry: ID=TYPE,Number=A,Type=String,Description="The type of
Warning: Expected at least 2 parts in INFO entry: ID=TYPE,Number=A,Type=String,Description="The type of
Warning: Expected at least 2 parts in INFO entry: ID=CIGAR,Number=A,Type=String,Description="The extend
Warning: Expected at least 2 parts in FORMAT entry: ID=GQ,Number=1,Type=Float,Description="Genotype Qua
Warning: Expected at least 2 parts in FORMAT entry: ID=GL,Number=G,Type=Float,Description="Genotype Lik
After filtering, kept 260 out of 260 Individuals
Outputting VCF file...
After filtering, kept 30 out of a possible 38 Sites
Run Time = 0.00 seconds

VCFtools - 0.1.17
(C) Adam Auton and Anthony Marcketta 2009

Parameters as interpreted:
    --vcf DQBex2_filt2.recode.vcf
    --missing-indv
    --out DQBex2_filt2

Warning: Expected at least 2 parts in INFO entry: ID=AF,Number=A,Type=Float,Description="Estimated alle
Warning: Expected at least 2 parts in INFO entry: ID=PRO,Number=1,Type=Float,Description="Reference all
Warning: Expected at least 2 parts in INFO entry: ID=PAO,Number=A,Type=Float,Description="Alternate all
Warning: Expected at least 2 parts in INFO entry: ID=SRP,Number=1,Type=Float,Description="Strand balanc
Warning: Expected at least 2 parts in INFO entry: ID=SAP,Number=A,Type=Float,Description="Strand balanc
Warning: Expected at least 2 parts in INFO entry: ID=AB,Number=A,Type=Float,Description="Allele balance
Warning: Expected at least 2 parts in INFO entry: ID=ABP,Number=A,Type=Float,Description="Allele balanc
Warning: Expected at least 2 parts in INFO entry: ID=RPP,Number=A,Type=Float,Description="Read Placemen
Warning: Expected at least 2 parts in INFO entry: ID=RPPR,Number=1,Type=Float,Description="Read Placeme
Warning: Expected at least 2 parts in INFO entry: ID=EPP,Number=A,Type=Float,Description="End Placement
Warning: Expected at least 2 parts in INFO entry: ID=EPPR,Number=1,Type=Float,Description="End Placemen
Warning: Expected at least 2 parts in INFO entry: ID=TYPE,Number=A,Type=String,Description="The type of
Warning: Expected at least 2 parts in INFO entry: ID=TYPE,Number=A,Type=String,Description="The type of
Warning: Expected at least 2 parts in INFO entry: ID=TYPE,Number=A,Type=String,Description="The type of
Warning: Expected at least 2 parts in INFO entry: ID=TYPE,Number=A,Type=String,Description="The type of
Warning: Expected at least 2 parts in INFO entry: ID=TYPE,Number=A,Type=String,Description="The type of
Warning: Expected at least 2 parts in INFO entry: ID=CIGAR,Number=A,Type=String,Description="The extend
Warning: Expected at least 2 parts in FORMAT entry: ID=GQ,Number=1,Type=Float,Description="Genotype Qua
Warning: Expected at least 2 parts in FORMAT entry: ID=GL,Number=G,Type=Float,Description="Genotype Lik
After filtering, kept 260 out of 260 Individuals
Outputting Individual Missingness
After filtering, kept 30 out of a possible 30 Sites
Run Time = 0.00 seconds
```

Identify individuals to remove based on missing data…

```r
ex2miss <- read.table("DQBex2_filt2.imiss", header = TRUE)
range(ex2miss$N_MISS)
```

```
## [1]  1 30
```

```r
hist(ex2miss$N_MISS, breaks=10)
```




```r
keepind <- ex2miss$INDV[which(ex2miss$N_MISS <= 5)]

write.table(keepind, "ex2_keepind.txt",
            quote = FALSE, row.names = FALSE, col.names = FALSE)
```

Keep only individuals with little missing data ($N_{MISS} \leq 5$)...

```
vcftools --vcf DQBex2_filt2.recode.vcf --out DQBex2_final \
          --keep ex2_keepind.txt --recode
```

```
VCFtools - 0.1.17
(C) Adam Auton and Anthony Marcketta 2009

Parameters as interpreted:
    --vcf DQBex2_filt2.recode.vcf
    --keep ex2_keepind.txt
    --out DQBex2_final
    --recode

Warning: Expected at least 2 parts in INFO entry: ID=AF,Number=A,Type=Float,Description="Estimated allel
Warning: Expected at least 2 parts in INFO entry: ID=PRO,Number=1,Type=Float,Description="Reference all
```

```
Warning: Expected at least 2 parts in INFO entry: ID=PAO,Number=A,Type=Float,Description="Alternate alle
Warning: Expected at least 2 parts in INFO entry: ID=SRP,Number=1,Type=Float,Description="Strand balanc
Warning: Expected at least 2 parts in INFO entry: ID=SAP,Number=A,Type=Float,Description="Strand balanc
Warning: Expected at least 2 parts in INFO entry: ID=AB,Number=A,Type=Float,Description="Allele balance
Warning: Expected at least 2 parts in INFO entry: ID=ABP,Number=A,Type=Float,Description="Allele balanc
Warning: Expected at least 2 parts in INFO entry: ID=RPP,Number=A,Type=Float,Description="Read Placemen
Warning: Expected at least 2 parts in INFO entry: ID=RPPR,Number=1,Type=Float,Description="Read Placeme
Warning: Expected at least 2 parts in INFO entry: ID=EPP,Number=A,Type=Float,Description="End Placement
Warning: Expected at least 2 parts in INFO entry: ID=EPPR,Number=1,Type=Float,Description="End Placemen
Warning: Expected at least 2 parts in INFO entry: ID=TYPE,Number=A,Type=String,Description="The type of
Warning: Expected at least 2 parts in INFO entry: ID=TYPE,Number=A,Type=String,Description="The type of
Warning: Expected at least 2 parts in INFO entry: ID=TYPE,Number=A,Type=String,Description="The type of
Warning: Expected at least 2 parts in INFO entry: ID=TYPE,Number=A,Type=String,Description="The type of
Warning: Expected at least 2 parts in INFO entry: ID=TYPE,Number=A,Type=String,Description="The type of
Warning: Expected at least 2 parts in INFO entry: ID=CIGAR,Number=A,Type=String,Description="The extend
Warning: Expected at least 2 parts in FORMAT entry: ID=GQ,Number=1,Type=Float,Description="Genotype Qua
Warning: Expected at least 2 parts in FORMAT entry: ID=GL,Number=G,Type=Float,Description="Genotype Lik
Keeping individuals in 'keep' list
After filtering, kept 195 out of 260 Individuals
Outputting VCF file...
After filtering, kept 30 out of a possible 30 Sites
Run Time = 0.00 seconds
```

Filtering *DQB* Exon 6

Filter for missing data, genotype quality, genotype depth, and so on…

```
#Site Quality, Genotyping Depth, Missingness, Minor Allele Count
vcftools --vcf DQBex6.sub.recode.vcf --out DQBex6_filt \
          --minQ 30 --minDP 10 --mac 2 \
          --remove-filtered-all --recode
```

```
VCFtools - 0.1.17
(C) Adam Auton and Anthony Marcketta 2009

Parameters as interpreted:
    --vcf DQBex6.sub.recode.vcf
    --mac 2
    --minDP 10
    --minQ 30
    --out DQBex6_filt
    --recode
    --remove-filtered-all

Warning: Expected at least 2 parts in INFO entry: ID=AF,Number=A,Type=Float,Description="Estimated allel
Warning: Expected at least 2 parts in INFO entry: ID=PRO,Number=1,Type=Float,Description="Reference all
Warning: Expected at least 2 parts in INFO entry: ID=PAO,Number=A,Type=Float,Description="Alternate all
Warning: Expected at least 2 parts in INFO entry: ID=SRP,Number=1,Type=Float,Description="Strand balanc
Warning: Expected at least 2 parts in INFO entry: ID=SAP,Number=A,Type=Float,Description="Strand balanc
Warning: Expected at least 2 parts in INFO entry: ID=AB,Number=A,Type=Float,Description="Allele balance
Warning: Expected at least 2 parts in INFO entry: ID=ABP,Number=A,Type=Float,Description="Allele balanc
Warning: Expected at least 2 parts in INFO entry: ID=RPP,Number=A,Type=Float,Description="Read Placemen
```

```
Warning: Expected at least 2 parts in INFO entry: ID=RPPR,Number=1,Type=Float,Description="Read Placemen
Warning: Expected at least 2 parts in INFO entry: ID=EPP,Number=A,Type=Float,Description="End Placement
Warning: Expected at least 2 parts in INFO entry: ID=EPPR,Number=1,Type=Float,Description="End Placemen
Warning: Expected at least 2 parts in INFO entry: ID=TYPE,Number=A,Type=String,Description="The type of
Warning: Expected at least 2 parts in INFO entry: ID=TYPE,Number=A,Type=String,Description="The type of
Warning: Expected at least 2 parts in INFO entry: ID=TYPE,Number=A,Type=String,Description="The type of
Warning: Expected at least 2 parts in INFO entry: ID=TYPE,Number=A,Type=String,Description="The type of
Warning: Expected at least 2 parts in INFO entry: ID=TYPE,Number=A,Type=String,Description="The type of
Warning: Expected at least 2 parts in INFO entry: ID=CIGAR,Number=A,Type=String,Description="The extend
Warning: Expected at least 2 parts in FORMAT entry: ID=GQ,Number=1,Type=Float,Description="Genotype Qua
Warning: Expected at least 2 parts in FORMAT entry: ID=GL,Number=G,Type=Float,Description="Genotype Lik
After filtering, kept 260 out of 260 Individuals
Outputting VCF file...
After filtering, kept 0 out of a possible 2 Sites
No data left for analysis!
Run Time = 0.00 seconds
```

This second region of *DQB* is largely composed of 3' UTR. The 2 SNPs that were not in the 3' UTR had low quality (both) or three alleles (one of the two). They do not pass our filters and we are left with only exon 2. See below for SNP qualities…

```
grep -v ^## DQBex6.sub.recode.vcf | cut -f 1,2,5,6
```

```
#CHROM  POS ALT QUAL
NW_025577149.1  1535177 A   0
NW_025577149.1  1535179 A,T 4.1659e-13
```

Phase *DQB* exon 2 SNPs with `beagle`

Now we use `beagle` to statistically phase the SNP loci in exon 2 of *DQB* and *impute genotypes* in samples with missing data. Since the phasing of SNPs involves imputation, we will only calculate haplotype-based measures of diversity from the phased dataset. For SNP-based diversity statistics, we'll return to the unphased dataset with missing data present.

```
beagle gt=DQBex2_final.recode.vcf out=DQBex2phased
```

```
beagle.27Feb25.75f.jar (version 5.5)
Copyright (C) 2014-2024 Brian L. Browning
Enter "java -jar beagle.27Feb25.75f.jar" to list command line argument
Start time: 10:51 AM EDT on 16 Jun 2026

Command line: java -Xmx989m -jar beagle.27Feb25.75f.jar
  gt=DQBex2_final.recode.vcf
  out=DQBex2phased
  nthreads=1

No genetic map is specified: using 1 cM = 1 Mb

Reference samples:                    0
Study     samples:                  195

Window 1 [NW_025577149.1:1531404-1531646]
```

```
Study     markers:                   30

Burnin  iteration 1:           0 seconds
Burnin  iteration 2:           0 seconds
Burnin  iteration 3:           0 seconds

Estimated ne:                  27899394
Estimated err:                 3.1e-03

Phasing iteration 1:           0 seconds
Phasing iteration 2:           0 seconds
Phasing iteration 3:           0 seconds
Phasing iteration 4:           0 seconds
Phasing iteration 5:           0 seconds
Phasing iteration 6:           0 seconds
Phasing iteration 7:           0 seconds
Phasing iteration 8:           0 seconds
Phasing iteration 9:           0 seconds
Phasing iteration 10:          0 seconds
Phasing iteration 11:          0 seconds
Phasing iteration 12:          0 seconds
WindWriter tot: 0 seconds

Cumulative Statistics:

Study     markers:                   30

Haplotype phasing time:        1 second
Total time:                    1 second

End time: 10:51 AM EDT on 16 Jun 2026
beagle.27Feb25.75f.jar finished
```

Re-annotate phased VCF

Run `snpEff` again with the phased dataset…

```
snpEff jax_bbear DQBex2phased.vcf.gz > DQBex2phased.ann.vcf
```

As a reminder, the annotation field has the following information…

```
grep ANN DQBex2phased.ann.vcf | head -1
```

```
##INFO=<ID=ANN,Number=.,Type=String,Description="Functional annotations: 'Allele | Annotation | Annotation_Impact | Gene_Name | Gene_ID | Feature_Type | Feature_ID | Transcript_BioType | Rank | HGVS.c | H
```

Here are the annotations for the exon 2 SNPs that pass our filters:

```
grep -v ^# DQBex2phased.ann.vcf | cut -f 1,2,8
```

```
NW_025577149.1  1531404 ANN=T|missense_variant|MODERATE|LOC123790950|LOC123790950|transcript|XM_045795559.1|protein_coding|2/6|c.106G>T|p.Val36Leu|218/1262|106/798|36/265||
NW_025577149.1  1531423 ANN=T|missense_variant|MODERATE|LOC123790950|LOC123790950|transcript|XM_045795559.1|protein_coding|2/6|c.125A>T|p.Glu42Val|237/1262|125/798|42/265||
NW_025577149.1  1531451 ANN=C|synonymous_variant|LOW|LOC123790950|LOC123790950|transcript|XM_045795559.1|protein_coding|2/6|c.153G>C|p.Arg51Arg|265/1262|153/798|51/265||
NW_025577149.1  1531454 ANN=A|synonymous_variant|LOW|LOC123790950|LOC123790950|transcript|XM_045795559.1|protein_coding|2/6|c.156G>A|p.Val52Val|268/1262|156/798|52/265||
NW_025577149.1  1531458 ANN=C|missense_variant|MODERATE|LOC123790950|LOC123790950|transcript|XM_045795559.1|protein_coding|2/6|c.160G>C|p.Gly54Arg|272/1262|160/798|54/265||
NW_025577149.1  1531459 ANN=T|missense_variant|MODERATE|LOC123790950|LOC123790950|transcript|XM_045795559.1|protein_coding|2/6|c.161G>T|p.Gly54Val|273/1262|161/798|54/265||
NW_025577149.1  1531464 ANN=G|missense_variant|MODERATE|LOC123790950|LOC123790950|transcript|XM_045795559.1|protein_coding|2/6|c.166A>G|p.Thr56Ala|278/1262|166/798|56/265||
NW_025577149.1  1531465 ANN=A|missense_variant|MODERATE|LOC123790950|LOC123790950|transcript|XM_045795559.1|protein_coding|2/6|c.167C>A|p.Thr56Asn|279/1262|167/798|56/265||
NW_025577149.1  1531470 ANN=T|missense_variant|MODERATE|LOC123790950|LOC123790950|transcript|XM_045795559.1|protein_coding|2/6|c.172C>T|p.His58Tyr|284/1262|172/798|58/265||
```

```
NW_025577149.1  1531471 ANN=C|missense_variant|MODERATE|LOC123790950|LOC123790950|transcript|XM_045795559.1|protein_coding|2/6|c.173A>C|p.His58Pro|285/1262|173/798|58/265||
NW_025577149.1  1531492 ANN=A|missense_variant|MODERATE|LOC123790950|LOC123790950|transcript|XM_045795559.1|protein_coding|2/6|c.194T>A|p.Phe65Tyr|306/1262|194/798|65/265||
NW_025577149.1  1531501 ANN=A|missense_variant|MODERATE|LOC123790950|LOC123790950|transcript|XM_045795559.1|protein_coding|2/6|c.203T>A|p.Phe68Tyr|315/1262|203/798|68/265||
NW_025577149.1  1531506 ANN=G|missense_variant|MODERATE|LOC123790950|LOC123790950|transcript|XM_045795559.1|protein_coding|2/6|c.208A>G|p.Ser70Gly|320/1262|208/798|70/265||
NW_025577149.1  1531507 ANN=A|missense_variant|MODERATE|LOC123790950|LOC123790950|transcript|XM_045795559.1|protein_coding|2/6|c.209G>A|p.Ser70Asn|321/1262|209/798|70/265||
NW_025577149.1  1531514 ANN=T|synonymous_variant|LOW|LOC123790950|LOC123790950|transcript|XM_045795559.1|protein_coding|2/6|c.216G>T|p.Val72Val|328/1262|216/798|72/265||
NW_025577149.1  1531521 ANN=T|missense_variant|MODERATE|LOC123790950|LOC123790950|transcript|XM_045795559.1|protein_coding|2/6|c.223C>T|p.His75Tyr|335/1262|223/798|75/265||
NW_025577149.1  1531525 ANN=A|missense_variant|MODERATE|LOC123790950|LOC123790950|transcript|XM_045795559.1|protein_coding|2/6|c.227G>A|p.Arg76Gln|339/1262|227/798|76/265||
NW_025577149.1  1531527 ANN=C|missense_variant|MODERATE|LOC123790950|LOC123790950|transcript|XM_045795559.1|protein_coding|2/6|c.229G>C|p.Ala77Pro|341/1262|229/798|77/265||
NW_025577149.1  1531535 ANN=C|synonymous_variant|LOW|LOC123790950|LOC123790950|transcript|XM_045795559.1|protein_coding|2/6|c.237G>C|p.Thr79Thr|349/1262|237/798|79/265||
NW_025577149.1  1531550 ANN=G|synonymous_variant|LOW|LOC123790950|LOC123790950|transcript|XM_045795559.1|protein_coding|2/6|c.252C>G|p.Pro84Pro|364/1262|252/798|84/265||
NW_025577149.1  1531556 ANN=C|synonymous_variant|LOW|LOC123790950|LOC123790950|transcript|XM_045795559.1|protein_coding|2/6|c.258T>C|p.Ala86Ala|370/1262|258/798|86/265||
NW_025577149.1  1531560 ANN=G|missense_variant|MODERATE|LOC123790950|LOC123790950|transcript|XM_045795559.1|protein_coding|2/6|c.262T>G|p.Tyr88Asp|374/1262|262/798|88/265||
NW_025577149.1  1531561 ANN=C|missense_variant|MODERATE|LOC123790950|LOC123790950|transcript|XM_045795559.1|protein_coding|2/6|c.263A>C|p.Tyr88Ser|375/1262|263/798|88/265||
NW_025577149.1  1531564 ANN=G|missense_variant|MODERATE|LOC123790950|LOC123790950|transcript|XM_045795559.1|protein_coding|2/6|c.266T>G|p.Leu89Trp|378/1262|266/798|89/265||
NW_025577149.1  1531565 ANN=C|missense_variant|MODERATE|LOC123790950|LOC123790950|transcript|XM_045795559.1|protein_coding|2/6|c.267G>C|p.Leu89Phe|379/1262|267/798|89/265||
NW_025577149.1  1531574 ANN=T|missense_variant|MODERATE|LOC123790950|LOC123790950|transcript|XM_045795559.1|protein_coding|2/6|c.276G>T|p.Gln92His|388/1262|276/798|92/265||
NW_025577149.1  1531591 ANN=G|missense_variant|MODERATE|LOC123790950|LOC123790950|transcript|XM_045795559.1|protein_coding|2/6|c.293A>G|p.Gln98Arg|405/1262|293/798|98/265||
NW_025577149.1  1531603 ANN=A|missense_variant|MODERATE|LOC123790950|LOC123790950|transcript|XM_045795559.1|protein_coding|2/6|c.305C>A|p.Ala102Glu|417/1262|305/798|102/265||
NW_025577149.1  1531612 ANN=A|missense_variant|MODERATE|LOC123790950|LOC123790950|transcript|XM_045795559.1|protein_coding|2/6|c.314C>A|p.Thr105Lys|426/1262|314/798|105/265||
NW_025577149.1  1531646 ANN=T|missense_variant|MODERATE|LOC123790950|LOC123790950|transcript|XM_045795559.1|protein_coding|2/6|c.348G>T|p.Arg116Ser|460/1262|348/798|116/265||
```

Now create a `.csv` file with annotations for the final set of SNPs…

```r
CPA <- data.frame(CHROM=system("grep -v ^# DQBex2phased.ann.vcf | cut -f 1",
                               intern = TRUE),
                  POS=system("grep -v ^# DQBex2phased.ann.vcf | cut -f 2",
                             intern = TRUE),
                  ANN=system("grep -v ^# DQBex2phased.ann.vcf | cut -f 8",
                             intern = TRUE))
write.csv(CPA, file="DQBex2ann.csv", quote = FALSE, row.names = FALSE)
```

Extract haplotype data from phased VCF

Use bcftools to extract haplotypes from the phased `.vcf` file…

```bash
bcftools convert --hapsample --vcf-ids DQBex2phased.ann.vcf -o DQBex2
gunzip DQBex2.hap.gz
```

```
Hap file: DQBex2.hap.gz
Sample file: DQBex2.samples
[W::vcf_parse] Contig 'NW_025577149.1' is not defined in the header. (Quick workaround: index the file w
30 records written, 0 skipped: 0/0/0 no-ALT/non-biallelic/filtered
```

Now read the output into R…

```r
#Read the sample IDs
sampIDs <- read.table("DQBex2.samples", header = TRUE)
sampIDs <- sampIDs[-1,]
dim(sampIDs)
```

```
## [1] 195   3
```

```r
#Read the hap data
hap <- read.table("DQBex2.hap", header = FALSE)
dim(hap)
```

```
## [1]  30 395
```

```
hap[1:10,1:10]
```

```
##               V1                          V2      V3 V4 V5 V6 V7 V8 V9 V10
## 1  NW_025577149.1 NW_025577149.1:1531404_G_T 1531404  G  T  0  0  0  0   0
## 2  NW_025577149.1 NW_025577149.1:1531423_A_T 1531423  A  T  0  0  0  0   0
## 3  NW_025577149.1 NW_025577149.1:1531451_G_C 1531451  G  C  0  0  0  0   0
## 4  NW_025577149.1 NW_025577149.1:1531454_G_A 1531454  G  A  0  0  0  0   0
## 5  NW_025577149.1 NW_025577149.1:1531458_G_C 1531458  G  C  1  1  1  1   1
## 6  NW_025577149.1 NW_025577149.1:1531459_G_T 1531459  G  T  1  1  1  1   1
## 7  NW_025577149.1 NW_025577149.1:1531464_A_G 1531464  A  G  0  0  0  0   0
## 8  NW_025577149.1 NW_025577149.1:1531465_C_A 1531465  C  A  0  0  0  0   0
## 9  NW_025577149.1 NW_025577149.1:1531470_C_T 1531470  C  T  1  1  1  1   1
## 10 NW_025577149.1 NW_025577149.1:1531471_A_C 1531471  A  C  0  0  0  0   0
```

```
#Name the columns
colnames(hap)[1:5] <- c("CHROM", "ID", "POS", "REF","ALT")
al1col <- seq(from=6, by=2, length.out=nrow(sampIDs))
for(i in 1:length(al1col)) {
  colnames(hap)[al1col[i]] <- paste0(sampIDs[i,1],".1")
  colnames(hap)[(al1col[i]+1)] <- paste0(sampIDs[i,1], ".2")
}

#Now join the genotypes into a haplotype string and save it to a data.frame
samps <- read.csv("../input/sampleMetaData_cloneID.csv", header = TRUE)
hap.out <- data.frame(vcfID=sampIDs[,1],
                      pop = NA, survivor=NA,
                      allele1 = NA, allele2 = NA)
for(i in 1:nrow(hap.out)) {
  hap.out$pop[i] <- samps$region[which(samps$vcfID == hap.out$vcfID[i])]
  hap.out$allele1[i] <- paste0(hap[,which(colnames(hap) ==
                                      paste0(hap.out$vcfID[i], ".1"))],
                               collapse="")
  hap.out$allele2[i] <- paste0(hap[,which(colnames(hap) ==
                                      paste0(hap.out$vcfID[i],".2"))],
                               collapse="")
  hap.out$survivor[i] <- ifelse(
    length(unique(
      samps$region[which(samps$finalClone ==
                     samps$finalClone[which(samps$vcfID ==
                                            hap.out$vcfID[i])])]
      )) == 1,
    FALSE, TRUE)
}
hap.out[1:10,]
```

```
##                  vcfID pop survivor                      allele1
## 1  UGA_173101_P020_WC08 CGA     TRUE 000011001010000100010001001100
## 2  UGA_173101_P020_WC01 CGA    FALSE 000011001010000100000111001100
## 3  UGA_173101_P020_WB12 CGA    FALSE 000011001000000100000111001101
## 4  UGA_173101_P020_WB10 CGA    FALSE 000100111000000100000111001101
## 5  UGA_173101_P020_WB05 CGA    FALSE 000011001000000100000111001101
## 6  UGA_173101_P019_WH08 NGA    FALSE 000011001010000100010001001100
```

```
## 7  UGA_173101_P019_WH06 NGA    FALSE 00010011101000010000011100110l
## 8  UGA_173101_P019_WH02 CGA    FALSE 000100111000000100000111001101
## 9  UGA_173101_P019_WG01 CGA    FALSE 000011001010000100000111001100
## 10 UGA_173101_P019_WF11 CGA    FALSE 000011001000000100000111001100
##                         allele2
## 1  000011001010000100010001001100
## 2  000011001000000100000111001101
## 3  000011001010000100000111001100
## 4  000100111010000100000111001101
## 5  000011001010000100000111001100
## 6  000011001010000100010001001100
## 7  000100111001110100000000100000
## 8  000100111000000100000111001101
## 9  000011001000000100000111001101
## 10 000011001010000100000111001100
```

Haplotype-based summaries

Haplotype richness (K), haplotype diversity ($\hat{H}$), and observed heterozygosity (H_o)

First, working with the number of unique haplotypes K and Nei's (1987) haplotype diversity, $\hat{H}$:

$$\hat{H} = \left(\frac{n}{n-1}\right)\left(1 - \sum_{i=1}^{k} x_i^2\right)$$

where n is the number of sequenced chromosomes, k is the number of unique haplotypes detected, and x_i is the frequency of the i^{th} haplotype.

```r
#Create population-specific objects
# For CGA23 and CGA12, don't forget to include surviving individuals!
CGA23haps <- hap.out[which(hap.out$pop == "CGA" | hap.out$survivor == TRUE),]
nrow(CGA23haps)
```

```
## [1] 135
```

```r
CGA12haps <- hap.out[which(hap.out$pop == "CGA-H" | hap.out$survivor == TRUE),]
nrow(CGA12haps)
```

```
## [1] 12
```

```r
NGAhaps <- hap.out[which(hap.out$pop == "NGA"),]
nrow(NGAhaps)
```

```
## [1] 26
```

```r
SGAhaps <- hap.out[which(hap.out$pop == "SGA"),]
nrow(SGAhaps)
```

```
## [1] 25
```

```r
#Number of unique haplotypes...
length(unique(c(hap.out$allele1, hap.out$allele2)))
```

```
## [1] 45
```

```r
length(unique(c(CGA23haps$allele1, CGA23haps$allele2)))
```

```
## [1] 30
```

```r
length(unique(c(CGA12haps$allele1, CGA12haps$allele2)))
```

```
## [1] 7
```

```r
length(unique(c(NGAhaps$allele1, NGAhaps$allele2)))
```

```
## [1] 19
```

```r
length(unique(c(SGAhaps$allele1, SGAhaps$allele2)))
```

```
## [1] 14
```

```r
#Haplotype diversity (Nei 1987)
hapdiv <- function(haps) {
  n <- 2*nrow(haps)
  haptab <- table(c(haps$allele1, haps$allele2))
  freqs <- haptab/sum(haptab)
  H <- (n/(n-1))*(1-sum(freqs^2))
  return(H)
}

#Nei's haplotype diversity
hapdiv(CGA23haps)
```

```
## [1] 0.861655
```

```r
hapdiv(CGA12haps)
```

```
## [1] 0.8478261
```

```r
hapdiv(NGAhaps)
```

```
## [1] 0.8695324
```

```r
hapdiv(SGAhaps)
```

```
## [1] 0.8612245
```

```r
#Observed heterozygosity (haplotype-based)
sum(CGA23haps$allele1 != CGA23haps$allele2)/nrow(CGA23haps)
```

```
## [1] 0.6592593
```

```r
sum(CGA12haps$allele1 != CGA12haps$allele2)/nrow(CGA12haps)
```

```
## [1] 0.4166667
```

```r
sum(NGAhaps$allele1 != NGAhaps$allele2)/nrow(NGAhaps)
```

```
## [1] 0.6153846
```

```r
sum(SGAhaps$allele1 != SGAhaps$allele2)/nrow(SGAhaps)
```

```
## [1] 0.64
```

Encouragingly, the central GA population seems to have retained quite a bit of diversity at exon 2 of *DQB*!

Haplotype-sharing across populations

We're also interested in how much differentiation there is in haplotype frequencies across the three black bear populations in GA.

```r
length(unique(c(hap.out$allele1, hap.out$allele2))) #Total unique haplotypes
```

```
## [1] 45
```

```r
pophapcnts <- data.frame(hapID=c(1:length(unique(c(hap.out$allele1,
                                                    hap.out$allele2)))),
                         DQBex2Hap=unique(c(hap.out$allele1, hap.out$allele2)),
                         NGA=NA,
                         CGA12=NA,
                         CGA23=NA,
                         SGA=NA)
for(haplo in pophapcnts$hapID) {
  #print(hap)
  #NGA frequencies
  pophapcnts$NGA[haplo] <-
    sum(c(hap.out$allele1[hap.out$pop=="NGA"],
          hap.out$allele2[hap.out$pop=="NGA"]) ==
          pophapcnts$DQBex2Hap[haplo])

  #SGA frequencies
  pophapcnts$SGA[haplo] <-
    sum(c(hap.out$allele1[hap.out$pop=="SGA"],
          hap.out$allele2[hap.out$pop=="SGA"]) ==
          pophapcnts$DQBex2Hap[haplo])
```

```r
  #CGA12 frequencies (CGA-H)
  pophapcnts$CGA12[haplo] <-
    sum(c(hap.out$allele1[hap.out$pop=="CGA-H"],
          hap.out$allele2[hap.out$pop=="CGA-H"]) ==
          pophapcnts$DQBex2Hap[haplo])

  #CGA23 frequencies (CGA)
  pophapcnts$CGA23[haplo] <-
    sum(c(hap.out$allele1[hap.out$pop=="CGA"],
          hap.out$allele2[hap.out$pop=="CGA"]) ==
          pophapcnts$DQBex2Hap[haplo])

}

#Haplotype counts
pophapcnts
```

```
##    hapID                      DQBex2Hap NGA CGA12 CGA23 SGA
## 1      1 000011001010000100010001001100  15     4    26  11
## 2      2 000011001010000100000111001100   1     3    40   4
## 3      3 000011001000000100000111001101   0     4    47   3
## 4      4 000100111000000100000111001101   1     4    66   0
## 5      5 000100111010000100000111001101   1     0     4   0
## 6      6 000011001000000100000111001100   2     0    13   5
## 7      7 000000000000000000010000100000   0     1     7   0
## 8      8 000011001010000100000101001100   0     0     2   0
## 9      9 000011001010000100000001001100   8     3    30  14
## 10    10 000011001010000100000000001100   2     0     0   1
## 11    11 000011001010000100000000101100   1     0     0   0
## 12    12 000011001010000100000011001100   0     0     3   0
## 13    13 000011001010000101000001001100   0     0     0   1
## 14    14 000100111001110100000111001100   1     0     0   0
## 15    15 000011001010000101001000100100   2     0     1   0
## 16    16 000011001010000100010000101100   1     0     4   1
## 17    17 000000000000000000010000100100   1     0     0   0
## 18    18 000100111001110100000000100000   9     0     2   2
## 19    19 000000000000000000000000000000   0     0     0   2
## 20    20 000100111000000100000111001100   2     0     1   0
## 21    21 000011000000000100000111001100   0     0     1   0
## 22    22 000011001010000101000000101100   1     0     0   0
## 23    23 000100111001000100000111001101   0     0     1   0
## 24    24 000100111010000100000111001100   0     0     1   0
## 25    25 000000011000000100000111001101   0     0     1   0
## 26    26 000000111001010000000111001100   0     0     1   0
## 27    27 000011001010000110100101010010   0     0     2   0
## 28    28 000100111100000100000111001101   0     0     2   0
## 29    29 111000101010001100110001000010   0     0     1   0
## 30    30 000011001010000000010000101100   0     0     1   0
## 31    31 000011001010000100000111001101   0     0     4   0
## 32    32 000000000000000100000111001100   0     0     2   0
## 33    33 000011001010000101000111001100   0     0     0   3
## 34    34 000011001010000100010000100100   0     1     0   1
## 35    35 000100111000000100000000100000   1     0     0   0
```

```
## 36    36 000100111010000101001111001100   0    0    1   0
## 37    37 000011001010000000000000000000   0    0    0   1
## 38    38 000011001010000100010001001101   1    0    0   0
## 39    39 000011001010000101010001001100   0    0    0   1
## 40    40 011000101010000100000111001100   1    0    0   0
## 41    41 011000101010001100100111000011   0    0    1   0
## 42    42 111000101010000100000111000011   0    0    1   0
## 43    43 000100111001110000000111001100   0    0    1   0
## 44    44 000011001010000100010000100000   1    0    0   0
## 45    45 000011001010000110110001001100   0    0    1   0
```

```
#Haplotype frequencies
pophapfreqs <- pophapcnts
for(popcol in c(3:6)) {
  pophapfreqs[,popcol] <- pophapfreqs[,popcol]/sum(pophapfreqs[,popcol])
}
pophapfreqs
```

```
##    hapID                      DQBex2Hap        NGA CGA12       CGA23  SGA
## 1      1 000011001010000100010001001100 0.28846154  0.20 0.097014925 0.22
## 2      2 000011001010000100000111001100 0.01923077  0.15 0.149253731 0.08
## 3      3 000011001000000100000111001101 0.00000000  0.20 0.175373134 0.06
## 4      4 000100111000000100000111001101 0.01923077  0.20 0.246268657 0.00
## 5      5 000100111010000100000111001101 0.01923077  0.00 0.014925373 0.00
## 6      6 000011001000000100000111001100 0.03846154  0.00 0.048507463 0.10
## 7      7 000000000000000000010000100000 0.00000000  0.05 0.026119403 0.00
## 8      8 000011001010000100000101001100 0.00000000  0.00 0.007462687 0.00
## 9      9 000011001010000100000001001100 0.15384615  0.15 0.111940299 0.28
## 10    10 000011001010000100000000001100 0.03846154  0.00 0.000000000 0.02
## 11    11 000011001010000100000000101100 0.01923077  0.00 0.000000000 0.00
## 12    12 000011001010000100000011001100 0.00000000  0.00 0.011194030 0.00
## 13    13 000011001010000101000001001100 0.00000000  0.00 0.000000000 0.02
## 14    14 000100111001110100000111001100 0.01923077  0.00 0.000000000 0.00
## 15    15 000011001010000101001000100100 0.03846154  0.00 0.003731343 0.00
## 16    16 000011001010000100010000101100 0.01923077  0.00 0.014925373 0.02
## 17    17 000000000000000000010000100100 0.01923077  0.00 0.000000000 0.00
## 18    18 000100111001110100000000100000 0.17307692  0.00 0.007462687 0.04
## 19    19 000000000000000000000000000000 0.00000000  0.00 0.000000000 0.04
## 20    20 000100111000000100000111001100 0.03846154  0.00 0.003731343 0.00
## 21    21 000011000000000100000111001100 0.00000000  0.00 0.003731343 0.00
## 22    22 000011001010000101000000101100 0.01923077  0.00 0.000000000 0.00
## 23    23 000100111001000100000111001101 0.00000000  0.00 0.003731343 0.00
## 24    24 000100111010000100000111001100 0.00000000  0.00 0.003731343 0.00
## 25    25 000000011000000100000111001101 0.00000000  0.00 0.003731343 0.00
## 26    26 000000111001010000000111001100 0.00000000  0.00 0.003731343 0.00
## 27    27 000011001010000110100101010010 0.00000000  0.00 0.007462687 0.00
## 28    28 000100111100000100000111001101 0.00000000  0.00 0.007462687 0.00
## 29    29 111000101010001100110001000010 0.00000000  0.00 0.003731343 0.00
## 30    30 000011001010000000010000101100 0.00000000  0.00 0.003731343 0.00
## 31    31 000011001010000100000111001101 0.00000000  0.00 0.014925373 0.00
## 32    32 000000000000000100000111001100 0.00000000  0.00 0.007462687 0.00
## 33    33 000011001010000101000111001100 0.00000000  0.00 0.000000000 0.06
## 34    34 000011001010000100010000100100 0.00000000  0.05 0.000000000 0.02
## 35    35 000100111000000100000000100000 0.01923077  0.00 0.000000000 0.00
```

```
## 36    36 000100111010000101001111001100 0.00000000  0.00 0.003731343 0.00
## 37    37 000011001010000000000000000000 0.00000000  0.00 0.000000000 0.02
## 38    38 000011001010000100010001001101 0.01923077  0.00 0.000000000 0.00
## 39    39 000011001010000101010001001100 0.00000000  0.00 0.000000000 0.02
## 40    40 011000101010000100000111001100 0.01923077  0.00 0.000000000 0.00
## 41    41 011000101010001100100111000011 0.00000000  0.00 0.003731343 0.00
## 42    42 111000101010000100000111000011 0.00000000  0.00 0.003731343 0.00
## 43    43 000100111001110000000111001100 0.00000000  0.00 0.003731343 0.00
## 44    44 000011001010000100010000100000 0.01923077  0.00 0.000000000 0.00
## 45    45 000011001010000110110001001100 0.00000000  0.00 0.003731343 0.00
```

```r
#Highest frequency haplotype in each population?
which.max(pophapfreqs[,3])
```

```
## [1] 1
```

```r
which.max(pophapfreqs[,4])
```

```
## [1] 1
```

```r
which.max(pophapfreqs[,5])
```

```
## [1] 4
```

```r
which.max(pophapfreqs[,6])
```

```
## [1] 9
```

```r
pophapfreqs[1:4,]
```

```
##   hapID                      DQBex2Hap        NGA CGA12      CGA23  SGA
## 1     1 000011001010000100010001001100 0.28846154  0.20 0.09701493 0.22
## 2     2 000011001010000100000111001100 0.01923077  0.15 0.14925373 0.08
## 3     3 000011001000000100000111001101 0.00000000  0.20 0.17537313 0.06
## 4     4 000100111000000100000111001101 0.01923077  0.20 0.24626866 0.00
```

There are some notable differences in the frequencies of haplotypes across the populations. For instance, the most frequent haplotype in both `NGA` and `SGA` (row #1 above) is at moderate frequency in `CGA23`.

Visualizing differences across populations

Here we'll take a look at the *proportion of each population* that carries each of the unique *DQB* alleles identified in our dataset. This is slightly different from a plot of haplotype frequencies by population and is inspired by Figure 3a of Grogan et al. (2017). This should help to illustrate patterns of diversity within populations and differentiation among populations at *DQB*.

```r
#Make an object that we can use to store the data more efficiently
# nrow is the number of individuals sampled total
# ncol is the number of unique haplotypes
DQBmat <- matrix(data = 0, nrow = nrow(hap.out), ncol = nrow(pophapcnts))
colnames(DQBmat) <- pophapcnts$hapID
for(i in 1:nrow(hap.out)) {
  hap.cols <- c(which(pophapcnts$DQBex2Hap == hap.out$allele1[i]),
                which(pophapcnts$DQBex2Hap == hap.out$allele2[i]))
  DQBmat[i,hap.cols[1]] <- DQBmat[i,hap.cols[1]]+1
  DQBmat[i,hap.cols[2]] <- DQBmat[i,hap.cols[2]]+1
  rm(hap.cols)
}
rowSums(DQBmat) #should all be 2 - 2 alleles per individual
```

```
##   [1] 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2
##  [38] 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2
##  [75] 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2
## [112] 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2
## [149] 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2
## [186] 2 2 2 2 2 2 2 2 2 2
```

```r
colSums(DQBmat) #should be equal to
```

```
##  1  2  3  4  5  6  7  8  9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25 26
## 56 48 54 71  5 20  8  2 55  3  1  3  1  1  3  6  1 13  2  3  1  1  1  1  1  1
## 27 28 29 30 31 32 33 34 35 36 37 38 39 40 41 42 43 44 45
##  2  2  1  1  4  2  3  2  1  1  1  1  1  1  1  1  1  1  1
```

```r
rowSums(pophapcnts[,-c(1,2)])
```

```
##  [1] 56 48 54 71  5 20  8  2 55  3  1  3  1  1  3  6  1 13  2  3  1  1  1  1  1
## [26]  1  2  2  1  1  4  2  3  2  1  1  1  1  1  1  1  1  1  1  1
```

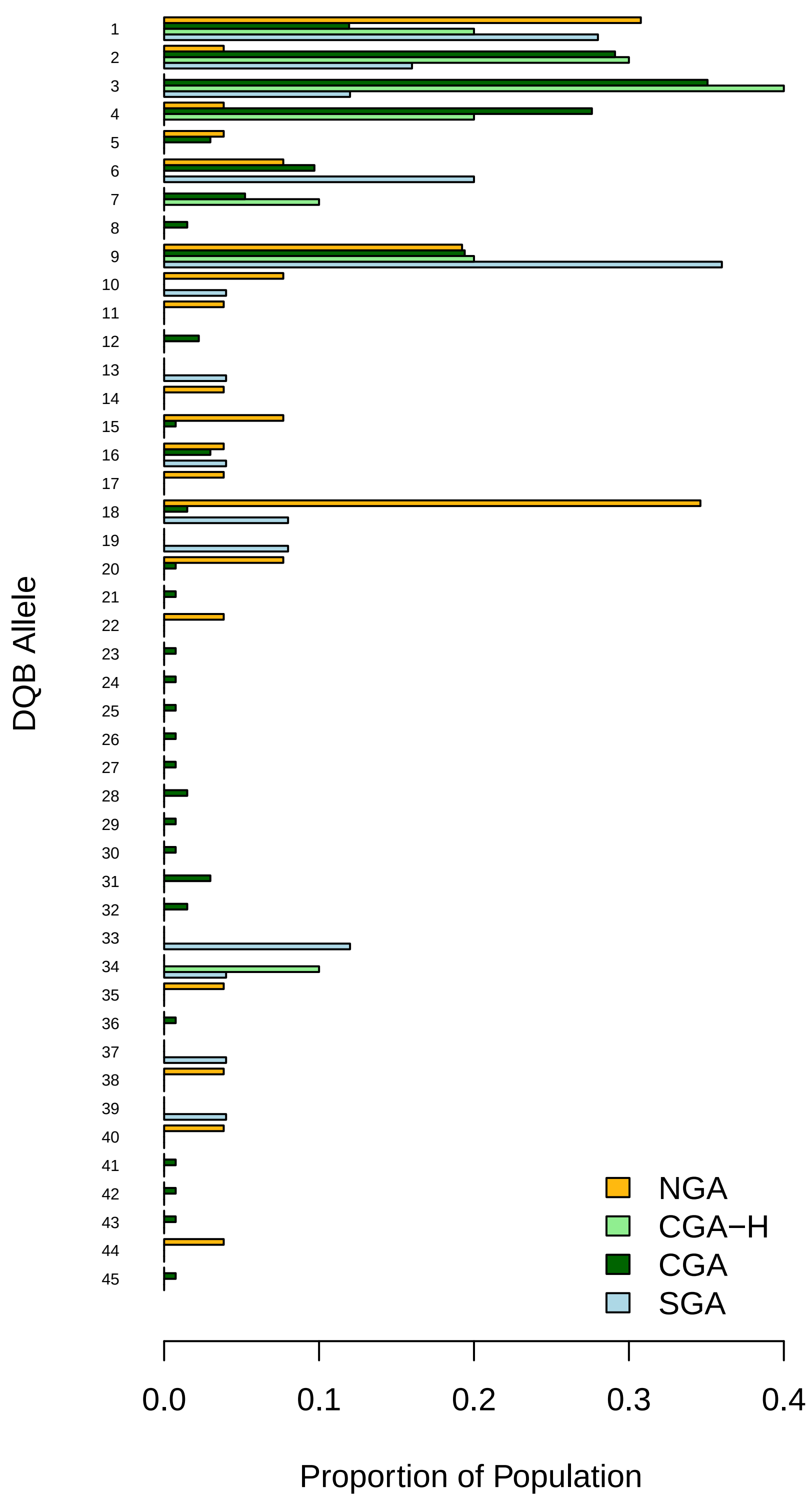
```r
sum(colSums(DQBmat) == rowSums(pophapcnts[,-c(1,2)])) #for all 44 unique haps
```

```
## [1] 45
```

```r
#Good!

#Now make an object with the proportion of the population that carries each hap
DQBprops <- data.frame(hapID = pophapcnts$hapID, NGA=0, CGA=0, CGAH=0, SGA=0)
for(i in 1:nrow(DQBprops)) {
  DQBprops$NGA[i] <- sum(DQBmat[hap.out$pop=="NGA",i]>0)/sum(hap.out$pop=="NGA")
  DQBprops$CGA[i] <- sum(DQBmat[hap.out$pop=="CGA",i]>0)/sum(hap.out$pop=="CGA")
  DQBprops$CGAH[i] <- sum(DQBmat[hap.out$pop=="CGA-H",i]>0)/sum(hap.out$pop=="CGA-H")
  DQBprops$SGA[i] <- sum(DQBmat[hap.out$pop=="SGA",i]>0)/sum(hap.out$pop=="SGA")
}
DQBprops <- t(DQBprops[,-1])

DQBprops
```

```
##            [,1]       [,2]      [,3]       [,4]       [,5]       [,6]       [,7]
## NGA  0.3076923 0.03846154 0.0000000 0.03846154 0.03846154 0.07692308 0.00000000
## CGA  0.1194030 0.29104478 0.3507463 0.27611940 0.02985075 0.09701493 0.05223881
## CGAH 0.2000000 0.30000000 0.4000000 0.20000000 0.00000000 0.00000000 0.10000000
## SGA  0.2800000 0.16000000 0.1200000 0.00000000 0.00000000 0.20000000 0.00000000
##             [,8]      [,9]      [,10]      [,11]      [,12] [,13]      [,14]
## NGA  0.00000000 0.1923077 0.07692308 0.03846154 0.00000000  0.00 0.03846154
## CGA  0.01492537 0.1940299 0.00000000 0.00000000 0.02238806  0.00 0.00000000
## CGAH 0.00000000 0.2000000 0.00000000 0.00000000 0.00000000  0.00 0.00000000
## SGA  0.00000000 0.3600000 0.04000000 0.00000000 0.00000000  0.04 0.00000000
##             [,15]      [,16]      [,17]      [,18] [,19]       [,20]       [,21]
## NGA  0.076923077 0.03846154 0.03846154 0.34615385  0.00 0.076923077 0.000000000
## CGA  0.007462687 0.02985075 0.00000000 0.01492537  0.00 0.007462687 0.007462687
## CGAH 0.000000000 0.00000000 0.00000000 0.00000000  0.00 0.000000000 0.000000000
## SGA  0.000000000 0.04000000 0.00000000 0.08000000  0.08 0.000000000 0.000000000
##           [,22]       [,23]       [,24]       [,25]       [,26]       [,27]
## NGA  0.03846154 0.000000000 0.000000000 0.000000000 0.000000000 0.000000000
## CGA  0.00000000 0.007462687 0.007462687 0.007462687 0.007462687 0.007462687
## CGAH 0.00000000 0.000000000 0.000000000 0.000000000 0.000000000 0.000000000
## SGA  0.00000000 0.000000000 0.000000000 0.000000000 0.000000000 0.000000000
##           [,28]       [,29]       [,30]      [,31]      [,32] [,33] [,34]
## NGA  0.00000000 0.000000000 0.000000000 0.00000000 0.00000000  0.00  0.00
## CGA  0.01492537 0.007462687 0.007462687 0.02985075 0.01492537  0.00  0.00
## CGAH 0.00000000 0.000000000 0.000000000 0.00000000 0.00000000  0.00  0.10
## SGA  0.00000000 0.000000000 0.000000000 0.00000000 0.00000000  0.12  0.04
##           [,35]       [,36] [,37]      [,38] [,39]      [,40]       [,41]
## NGA  0.03846154 0.000000000  0.00 0.03846154  0.00 0.03846154 0.000000000
## CGA  0.00000000 0.007462687  0.00 0.00000000  0.00 0.00000000 0.007462687
## CGAH 0.00000000 0.000000000  0.00 0.00000000  0.00 0.00000000 0.000000000
## SGA  0.00000000 0.000000000  0.04 0.00000000  0.04 0.00000000 0.000000000
##            [,42]       [,43]      [,44]       [,45]
## NGA  0.000000000 0.000000000 0.03846154 0.000000000
## CGA  0.007462687 0.007462687 0.00000000 0.007462687
## CGAH 0.000000000 0.000000000 0.00000000 0.000000000
## SGA  0.000000000 0.000000000 0.00000000 0.000000000
```

```
barplot(DQBprops[c(4,3,2,1),ncol(DQBprops):1], beside=TRUE, horiz=TRUE,
        names = c(ncol(DQBprops):1), las=1, cex.names=0.5,
        xlab = "Proportion of Population", ylab = "DQB Allele",
        col = c("lightblue","lightgreen","darkgreen","darkgoldenrod1"))
legend("bottomright", bty="n",
       fill=c("darkgoldenrod1","lightgreen","darkgreen","lightblue"),
       legend = c("NGA","CGA-H","CGA","SGA"))
```


DQB Allele
1
2
3
4
5
6
7
8
9
10
11
12
13
14
15
16
17
18
19
20
21
22
23
24
25
26
27
28
29
30
31
32
33
34
35
36
37
38
39
40
41
42
43
44
45
NGA
CGA−H
CGA
SGA
0.0
0.1
0.2
0.3
0.4
Proportion of Population

Influences of sample size

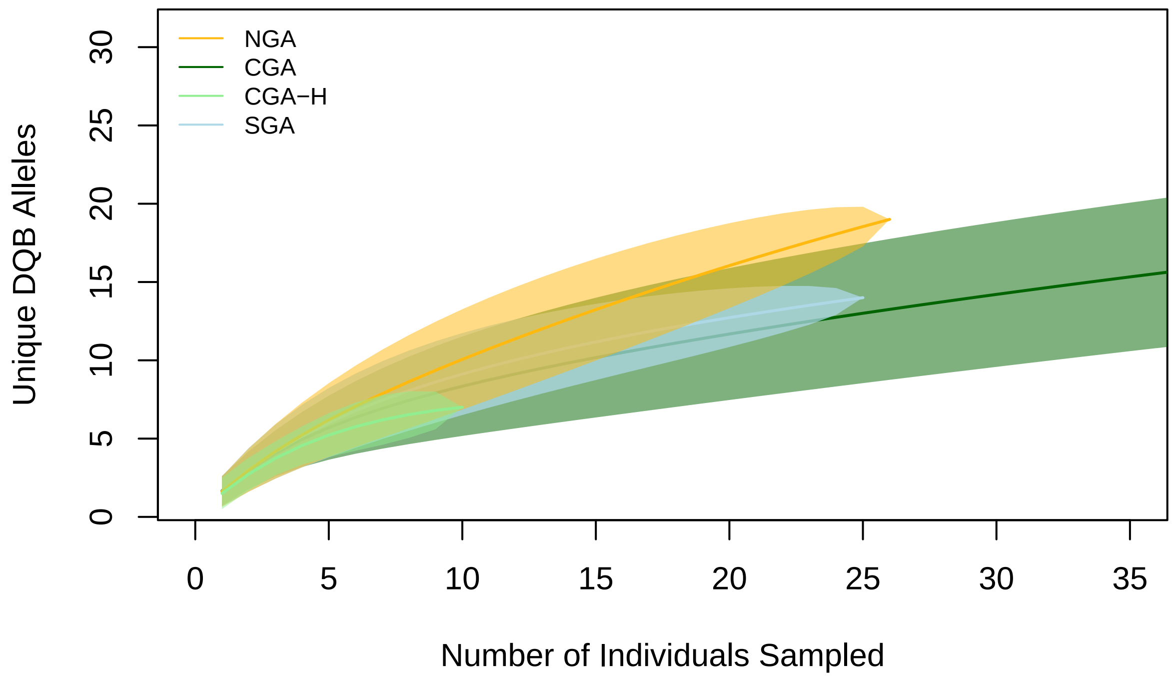
The imbalanced sample sizes across populations makes it difficult to compare diversity levels in plots like the one above. Grogan et al. (2013) used rarefaction approaches (similar to species accumuation curves) to compare the accumulation of unique alleles with additional samples in captive and wild populations of ring-tailed lemurs (*Lemur catta*). We'll use the `specaccum` function from the R package `vegan` (with `method = "exact"`) to do something similar here. See the "Vegan: ecological diversity" vignette (section 5.1, in particular) for methods that underlie `specaccum`. Kindt's exact method is the recommended approach for species accumulation curves, thus we use that approach below.

The `specaccum` function takes a community dataset where rows are site visits (in our case, sampled individuals) and columns are species (alleles, in our case). So we can use the object created above (`DQBmat`) for this analysis. Note that we are leaving out the 2012 samples from the central GA population. The numbers there are less than half the sample size in the next most poorly sampled population, including that sample would unnecessarily constrain the accumulation curves for the comparisons of interest.

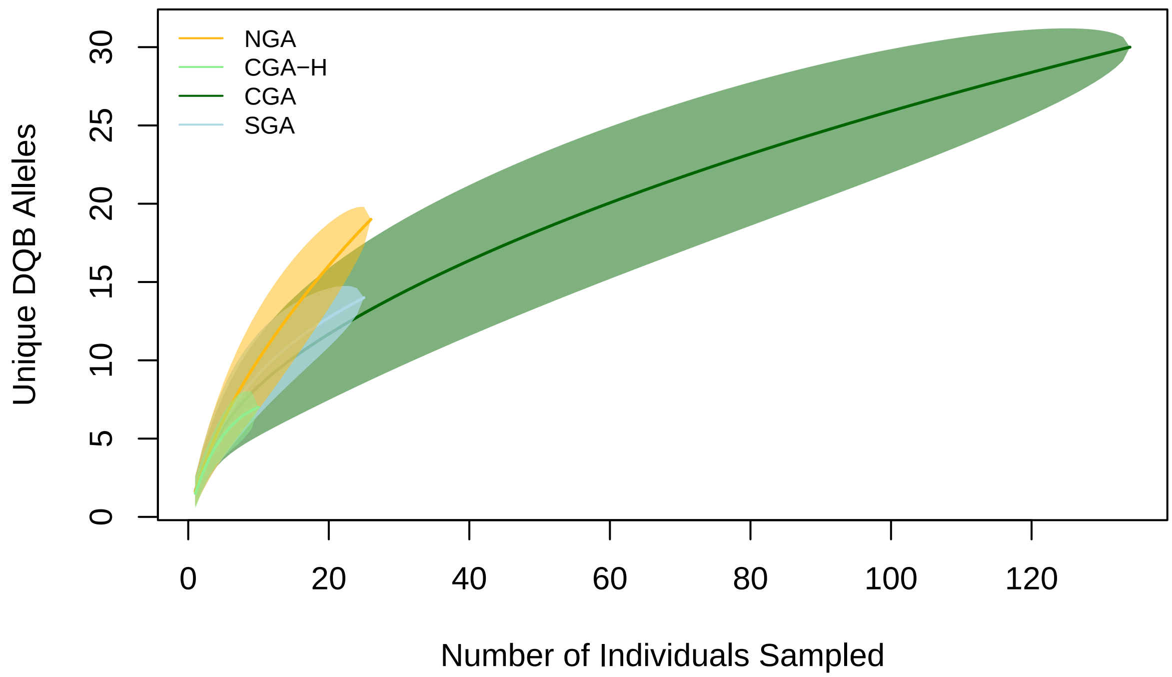
```
library(vegan)
```

```
## Loading required package: permute
```

```
library(scales)

NGAmat <- DQBmat[which(hap.out$pop == "NGA"),]
CGAmat <- DQBmat[which(hap.out$pop == "CGA"),]
SGAmat <- DQBmat[which(hap.out$pop == "SGA"),]
CGAHmat <- DQBmat[which(hap.out$pop == "CGA-H"),]

NGAsac <- specaccum(NGAmat, method = "exact")
SGAsac <- specaccum(SGAmat, method = "exact")
CGAsac <- specaccum(CGAmat, method = "exact")
CGAHsac <- specaccum(CGAHmat, method = "exact")

#Truncate the plot to avoid plotting very large n in CGA 2023
plot(CGAsac, xlim = c(0,35), ci.type="polygon", lwd = 1.5,
     col = "darkgreen", ci.col = alpha("darkgreen", 0.5), ci.lty=0,
     xlab = "Number of Individuals Sampled", ylab = "Unique DQB Alleles")
plot(SGAsac, add = TRUE, ci.type = "polygon", lwd = 1.5,
     col = "lightblue", ci.col = alpha("lightblue", 0.75), ci.lty=0)
plot(NGAsac, add = TRUE, ci.type = "polygon", lwd = 1.5,
     col = "darkgoldenrod1", ci.col=alpha("darkgoldenrod1", 0.5), ci.lty=0)
plot(CGAHsac, add = TRUE, ci.type = "polygon", lwd = 1.5,
     col = "lightgreen", ci.col=alpha("lightgreen",0.5), ci.lty=0)
legend("topleft", lty = 1, bty = "n", cex = 0.75,
       col = c("darkgoldenrod1","darkgreen","lightgreen","lightblue"),
       legend = c("NGA","CGA","CGA-H","SGA"))
```




```r
#Same plot, without truncation
plot(CGAsac, ci.type="polygon", lwd = 1.5,
     col = "darkgreen", ci.col = alpha("darkgreen", 0.5), ci.lty=0,
     xlab = "Number of Individuals Sampled", ylab = "Unique DQB Alleles")
plot(SGAsac, add = TRUE, ci.type = "polygon", lwd = 1.5,
     col = "lightblue", ci.col = alpha("lightblue", 0.75), ci.lty=0)
plot(NGAsac, add = TRUE, ci.type = "polygon", lwd = 1.5,
     col = "darkgoldenrod1", ci.col=alpha("darkgoldenrod1", 0.5), ci.lty=0)
plot(CGAHsac, add = TRUE, ci.type = "polygon", lwd = 1.5,
     col = "lightgreen", ci.col=alpha("lightgreen",0.5), ci.lty=0)
legend("topleft", lty = 1, bty = "n", cex = 0.75,
       col = c("darkgoldenrod1","lightgreen","darkgreen","lightblue"),
       legend = c("NGA","CGA-H","CGA","SGA"))
```

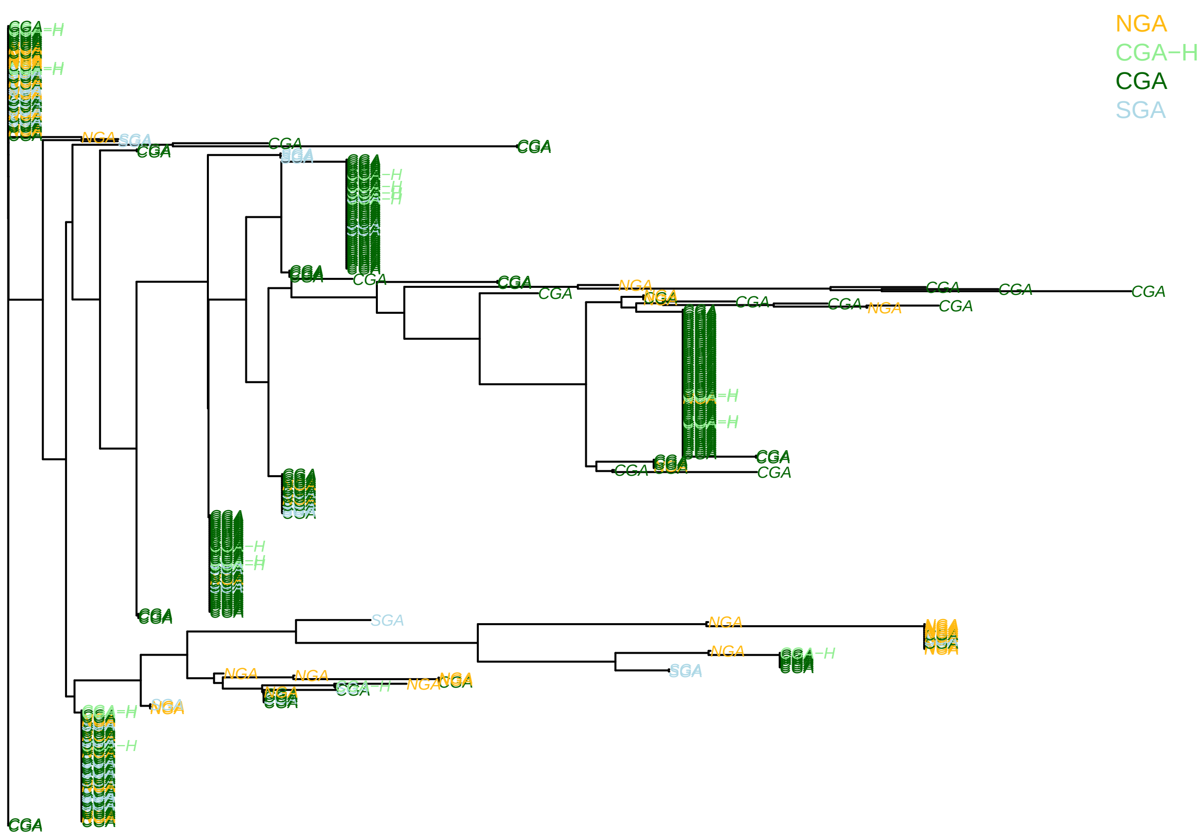


Neighbor-Joining Tree of *DQB*

Now we use the package `ape` to make a neighbor-joining phylogeny of *DQB*. Note that, since this tree is based on SNP data only, the branch lengths are a bit exaggerated (*i.e.*, invariant sites are not included).

```
#First, convert to a fake DNA sequence - gsub 0/1 to REF/ALT
seqhap <- hap[,-c(1:5)]
for(snp in 1:nrow(hap)) {
  seqhap[snp,] <- gsub("0", hap$REF[snp], seqhap[snp,])
  seqhap[snp,] <- gsub("1", hap$ALT[snp], seqhap[snp,])
}

library(ape)
DQBaln <- as.alignment(seqhap)
DQBbin <- as.DNAbin(DQBaln)
dmat <- dist.dna(DQBbin, model = "N")
NJtree <- nj(dmat)
#Relabel tips as pops
poplabels <- tipcols <- c()
for(i in 1:length(NJtree$tip.label)) {
  poplabels[i] <- hap.out$pop[which(hap.out$vcfID ==
                                      strsplit(NJtree$tip.label[i],
                                               split=".",
                                               fixed=TRUE)[[1]][1])]
}
```

```r
NJtree$tip.label <- poplabels

tipcols[which(poplabels == "NGA")] <- "darkgoldenrod1"
tipcols[which(poplabels == "SGA")] <- "lightblue"
tipcols[which(poplabels == "CGA")] <- "darkgreen"
tipcols[which(poplabels == "CGA-H")] <- "lightgreen"

#Plot the tree
par(mar=c(0,0,0,0))
plot(NJtree, tip.col=tipcols, cex=0.5)
legend("topright", bty = "n", cex = 0.75,
       legend = c("NGA","CGA-H","CGA","SGA"),
       text.col=c("darkgoldenrod1","lightgreen","darkgreen","lightblue"))
```



Haplotype network of *DQB*

Now a haplotype network using tools from the `pegas` package. The below code closely follows the vignette from the `pegas` package [Plotting Haplotype Networks with `pegas`](#).

```r
library(pegas)
```

```
##
## Attaching package: 'pegas'
```

```
## The following object is masked from 'package:ape':
##
##     mst
```

```
library(network)
```

```
##
## 'network' 1.19.0 (2024-12-08), part of the Statnet Project
## * 'news(package="network")' for changes since last version
## * 'citation("network")' for citation information
## * 'https://statnet.org' for help, support, and other information
```

```
#Extract unique haplotypes
uniq.haps <- haplotype(DQBbin)
uniq.haps
```

```
##
## Haplotypes extracted from: DQBbin
##
##     Number of haplotypes: 45
##          Sequence length: 30
##
## Haplotype labels and frequencies:
##
##       I      II     III      IV       V      VI     VII    VIII      IX       X
##      56      48      54      71       5      13      20       1       8       1
##      XI     XII    XIII     XIV      XV     XVI    XVII   XVIII     XIX      XX
##       2      55       3       3       4       1       2       3       1       3
##     XXI    XXII   XXIII    XXIV     XXV    XXVI   XXVII  XXVIII    XXIX     XXX
##       2       6       1       1       3       1       1       2       1       1
##    XXXI   XXXII  XXXIII   XXXIV    XXXV   XXXVI  XXXVII XXXVIII   XXXIX      XL
##       1       1       1       1       1       1       1       1       1       1
## ...
## (use summary() to print all)
```

```
#Distances among haplotypes - # of bp differences
hap.dist <- dist.dna(uniq.haps, model="N")
#Randomized minimum spanning tree for the haplotype network
hapnet <- rmst(hap.dist)
hapnet
```

```
## Haplotype network with:
##   45 haplotypes
##   55 links
##   link lengths between 1 and 6 step(s)
##
## Use print.default() to display all elements.
```

```
#Record the counts of each haplotype for the plot
sz <- summary(uniq.haps)
sz
```

```
##       I      II     III      IV       V      VI     VII    VIII      IX       X
##      56      48      54      71       5      13      20       1       8       1
##      XI     XII    XIII     XIV      XV     XVI    XVII   XVIII     XIX      XX
##       2      55       3       3       4       1       2       3       1       3
##     XXI    XXII   XXIII    XXIV     XXV    XXVI   XXVII  XXVIII    XXIX     XXX
##       2       6       1       1       3       1       1       2       1       1
##    XXXI   XXXII  XXXIII   XXXIV    XXXV   XXXVI  XXXVII XXXVIII   XXXIX      XL
##       1       1       1       1       1       1       1       1       1       1
##     XLI    XLII   XLIII    XLIV     XLV
##       1       2       2       1       1
```

```
hapnet.labels <- attr(hapnet,"labels")
hapnet.labels
```

```
##  [1] "I"        "II"       "III"      "IV"       "V"        "VI"       "VII"
##  [8] "VIII"     "IX"       "X"        "XI"       "XII"      "XIII"     "XIV"
## [15] "XV"       "XVI"      "XVII"     "XVIII"    "XIX"      "XX"       "XXI"
## [22] "XXII"     "XXIII"    "XXIV"     "XXV"      "XXVI"     "XXVII"    "XXVIII"
## [29] "XXIX"     "XXX"      "XXXI"     "XXXII"    "XXXIII"   "XXXIV"    "XXXV"
## [36] "XXXVI"    "XXXVII"   "XXXVIII" "XXXIX"     "XL"       "XLI"      "XLII"
## [43] "XLIII"    "XLIV"     "XLV"
```

```
sz <- sz[hapnet.labels]
sz
```

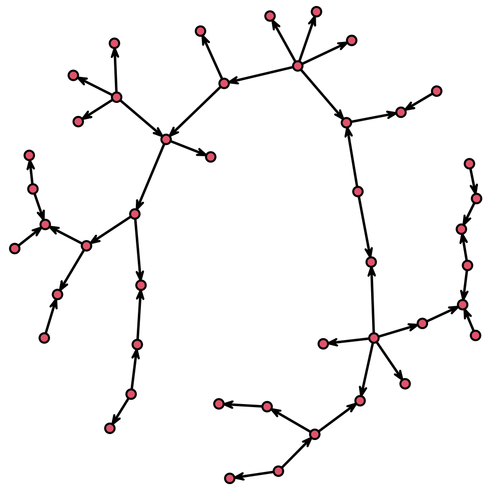
```
##       I      II     III      IV       V      VI     VII    VIII      IX       X
##      56      48      54      71       5      13      20       1       8       1
##      XI     XII    XIII     XIV      XV     XVI    XVII   XVIII     XIX      XX
##       2      55       3       3       4       1       2       3       1       3
##     XXI    XXII   XXIII    XXIV     XXV    XXVI   XXVII  XXVIII    XXIX     XXX
##       2       6       1       1       3       1       1       2       1       1
##    XXXI   XXXII  XXXIII   XXXIV    XXXV   XXXVI  XXXVII XXXVIII   XXXIX      XL
##       1       1       1       1       1       1       1       1       1       1
##     XLI    XLII   XLIII    XLIV     XLV
##       1       2       2       1       1
```

```
pie.size <- sqrt(sz/pi)
#Scale by sqrt(count / pi) so AREA of circles are proportional to numbers

#Compute frequencies of haplotypes for each population
pop.freq <- haploFreq(DQBbin, fac=poplabels, haplo=uniq.haps)
pop.freq <- pop.freq[hapnet.labels,]
pop.freq
```

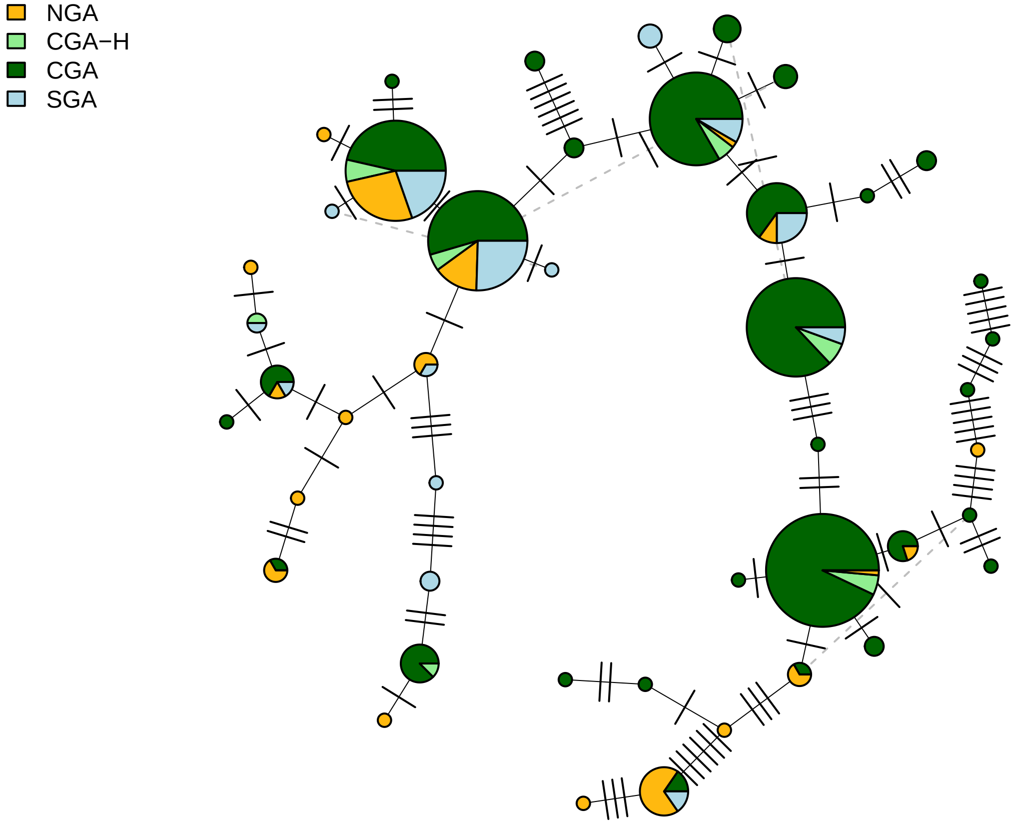
```
##         CGA CGA-H NGA SGA
## I        26     4  15  11
## II       40     3   1   4
## III      47     4   0   3
## IV       66     4   1   0
## V         4     0   1   0
## VI        2     0   9   2
## VII      13     0   2   5
```

```
## VIII      1   0   0   0
## IX        7   1   0   0
## X         1   0   0   0
## XI        2   0   0   0
## XII      30   3   8  14
## XIII      3   0   0   0
## XIV       0   0   2   1
## XV        4   0   0   0
## XVI       0   0   1   0
## XVII      2   0   0   0
## XVIII     0   0   0   3
## XIX       0   0   0   1
## XX        1   0   2   0
## XXI       0   1   0   1
## XXII      4   0   1   1
## XXIII     0   0   1   0
## XXIV      0   0   1   0
## XXV       1   0   2   0
## XXVI      0   0   1   0
## XXVII     1   0   0   0
## XXVIII    0   0   0   2
## XXIX      0   0   0   1
## XXX       0   0   1   0
## XXXI      0   0   0   1
## XXXII     0   0   1   0
## XXXIII    1   0   0   0
## XXXIV     1   0   0   0
## XXXV      0   0   1   0
## XXXVI     1   0   0   0
## XXXVII    1   0   0   0
## XXXVIII   1   0   0   0
## XXXIX     1   0   0   0
## XL        1   0   0   0
## XLI       1   0   0   0
## XLII      2   0   0   0
## XLIII     2   0   0   0
## XLIV      0   0   1   0
## XLV       1   0   0   0
```

```
#Use the network package to establish a reasonable layout for the network
# this solution is from the help for ?plot.haploNet & this post:
# https://stat.ethz.ch/pipermail/r-sig-genetics/2022-July/000237.html
netxy <- plot(as.network(hapnet))
```



```
xy <- list(x=netxy[,1], y=netxy[,2])

par(mar=c(0,0,0,0))
plot(hapnet, size=pie.size, xy=xy, pie=pop.freq,
     labels=FALSE, show.mutation=1, lwd=0.5,
     bg=c("darkgreen","lightgreen","darkgoldenrod1","lightblue"))
legend("topleft", bty="n", cex = 0.75,
       fill=c("darkgoldenrod1","lightgreen","darkgreen","lightblue"),
       legend=c("NGA","CGA-H","CGA","SGA"))
```



Predicted effects of mutations

Given the filters applied above and the lack of sequence data for exon 2 of the *DQB* gene, we will not attempt to calculate a $\frac{d_N}{d_S}$ ratio from the `snpEff` annotation. Methods to calculate these ratios typically require aligned sequences, rather than a `.vcf` file with only SNP loci. While the `snpEff` annotation does tell us about the number of non-synonymous substitutions, it does not tell us about the possible number of mutations at each site that would have led to synonymous and non-synonymous changes. Thus, we will focus on a simple summary of the numbers of synonymous and non-synonymous mutations that underlie the polymorphic SNP loci in each population.

First, create `.vcf` files for each population.

```
write.table(CGA12haps$vcfID, file = "CGA12mhckeep.txt",
            quote = FALSE,
            row.names = FALSE,
            col.names = FALSE)
write.table(CGA23haps$vcfID, file = "CGA23mhckeep.txt",
            quote = FALSE,
            row.names = FALSE,
            col.names = FALSE)
write.table(NGAhaps$vcfID, file = "NGAmhckeep.txt",
            quote = FALSE,
            row.names = FALSE,
            col.names = FALSE)
write.table(SGAhaps$vcfID, file = "SGAmhckeep.txt",
```

```
            quote = FALSE,
            row.names = FALSE,
            col.names = FALSE)
```

Now extract each population into its own `.vcf` file and rename the historic samples so survivors are present in both 2012 and 2023 populations…

```
vcftools --vcf DQBex2_final.recode.vcf --keep CGA12mhckeep.txt \
          --recode --out DQBex2_CGA12
vcftools --vcf DQBex2_final.recode.vcf --keep CGA23mhckeep.txt \
          --recode --out DQBex2_CGA23
vcftools --vcf DQBex2_final.recode.vcf --keep NGAmhckeep.txt \
          --recode --out DQBex2_NGA
vcftools --vcf DQBex2_final.recode.vcf --keep SGAmhckeep.txt \
          --recode --out DQBex2_SGA
```

```
VCFtools - 0.1.17
(C) Adam Auton and Anthony Marcketta 2009

Parameters as interpreted:
    --vcf DQBex2_final.recode.vcf
    --keep CGA12mhckeep.txt
    --out DQBex2_CGA12
    --recode

Warning: Expected at least 2 parts in INFO entry: ID=AF,Number=A,Type=Float,Description="Estimated allel
Warning: Expected at least 2 parts in INFO entry: ID=PRO,Number=1,Type=Float,Description="Reference allo
Warning: Expected at least 2 parts in INFO entry: ID=PAO,Number=A,Type=Float,Description="Alternate allo
Warning: Expected at least 2 parts in INFO entry: ID=SRP,Number=1,Type=Float,Description="Strand balance
Warning: Expected at least 2 parts in INFO entry: ID=SAP,Number=A,Type=Float,Description="Strand balance
Warning: Expected at least 2 parts in INFO entry: ID=AB,Number=A,Type=Float,Description="Allele balance
Warning: Expected at least 2 parts in INFO entry: ID=ABP,Number=A,Type=Float,Description="Allele balance
Warning: Expected at least 2 parts in INFO entry: ID=RPP,Number=A,Type=Float,Description="Read Placement
Warning: Expected at least 2 parts in INFO entry: ID=RPPR,Number=1,Type=Float,Description="Read Placemen
Warning: Expected at least 2 parts in INFO entry: ID=EPP,Number=A,Type=Float,Description="End Placement
Warning: Expected at least 2 parts in INFO entry: ID=EPPR,Number=1,Type=Float,Description="End Placement
Warning: Expected at least 2 parts in INFO entry: ID=TYPE,Number=A,Type=String,Description="The type of
Warning: Expected at least 2 parts in INFO entry: ID=TYPE,Number=A,Type=String,Description="The type of
Warning: Expected at least 2 parts in INFO entry: ID=TYPE,Number=A,Type=String,Description="The type of
Warning: Expected at least 2 parts in INFO entry: ID=TYPE,Number=A,Type=String,Description="The type of
Warning: Expected at least 2 parts in INFO entry: ID=TYPE,Number=A,Type=String,Description="The type of
Warning: Expected at least 2 parts in INFO entry: ID=CIGAR,Number=A,Type=String,Description="The extende
Warning: Expected at least 2 parts in FORMAT entry: ID=GQ,Number=1,Type=Float,Description="Genotype Qual
Warning: Expected at least 2 parts in FORMAT entry: ID=GL,Number=G,Type=Float,Description="Genotype Lik
Keeping individuals in 'keep' list
After filtering, kept 12 out of 195 Individuals
Outputting VCF file...
After filtering, kept 30 out of a possible 30 Sites
Run Time = 0.00 seconds

VCFtools - 0.1.17
(C) Adam Auton and Anthony Marcketta 2009
```

```
Parameters as interpreted:
    --vcf DQBex2_final.recode.vcf
    --keep CGA23mhckeep.txt
    --out DQBex2_CGA23
    --recode

Warning: Expected at least 2 parts in INFO entry: ID=AF,Number=A,Type=Float,Description="Estimated allel
Warning: Expected at least 2 parts in INFO entry: ID=PRO,Number=1,Type=Float,Description="Reference allo
Warning: Expected at least 2 parts in INFO entry: ID=PAO,Number=A,Type=Float,Description="Alternate allo
Warning: Expected at least 2 parts in INFO entry: ID=SRP,Number=1,Type=Float,Description="Strand balance
Warning: Expected at least 2 parts in INFO entry: ID=SAP,Number=A,Type=Float,Description="Strand balance
Warning: Expected at least 2 parts in INFO entry: ID=AB,Number=A,Type=Float,Description="Allele balance
Warning: Expected at least 2 parts in INFO entry: ID=ABP,Number=A,Type=Float,Description="Allele balance
Warning: Expected at least 2 parts in INFO entry: ID=RPP,Number=A,Type=Float,Description="Read Placement
Warning: Expected at least 2 parts in INFO entry: ID=RPPR,Number=1,Type=Float,Description="Read Placemen
Warning: Expected at least 2 parts in INFO entry: ID=EPP,Number=A,Type=Float,Description="End Placement
Warning: Expected at least 2 parts in INFO entry: ID=EPPR,Number=1,Type=Float,Description="End Placement
Warning: Expected at least 2 parts in INFO entry: ID=TYPE,Number=A,Type=String,Description="The type of
Warning: Expected at least 2 parts in INFO entry: ID=TYPE,Number=A,Type=String,Description="The type of
Warning: Expected at least 2 parts in INFO entry: ID=TYPE,Number=A,Type=String,Description="The type of
Warning: Expected at least 2 parts in INFO entry: ID=TYPE,Number=A,Type=String,Description="The type of
Warning: Expected at least 2 parts in INFO entry: ID=TYPE,Number=A,Type=String,Description="The type of
Warning: Expected at least 2 parts in INFO entry: ID=CIGAR,Number=A,Type=String,Description="The extende
Warning: Expected at least 2 parts in FORMAT entry: ID=GQ,Number=1,Type=Float,Description="Genotype Qual
Warning: Expected at least 2 parts in FORMAT entry: ID=GL,Number=G,Type=Float,Description="Genotype Like
Keeping individuals in 'keep' list
After filtering, kept 135 out of 195 Individuals
Outputting VCF file...
After filtering, kept 30 out of a possible 30 Sites
Run Time = 0.00 seconds

VCFtools - 0.1.17
(C) Adam Auton and Anthony Marcketta 2009

Parameters as interpreted:
    --vcf DQBex2_final.recode.vcf
    --keep NGAmhckeep.txt
    --out DQBex2_NGA
    --recode

Warning: Expected at least 2 parts in INFO entry: ID=AF,Number=A,Type=Float,Description="Estimated allel
Warning: Expected at least 2 parts in INFO entry: ID=PRO,Number=1,Type=Float,Description="Reference allo
Warning: Expected at least 2 parts in INFO entry: ID=PAO,Number=A,Type=Float,Description="Alternate allo
Warning: Expected at least 2 parts in INFO entry: ID=SRP,Number=1,Type=Float,Description="Strand balance
Warning: Expected at least 2 parts in INFO entry: ID=SAP,Number=A,Type=Float,Description="Strand balance
Warning: Expected at least 2 parts in INFO entry: ID=AB,Number=A,Type=Float,Description="Allele balance
Warning: Expected at least 2 parts in INFO entry: ID=ABP,Number=A,Type=Float,Description="Allele balance
Warning: Expected at least 2 parts in INFO entry: ID=RPP,Number=A,Type=Float,Description="Read Placement
Warning: Expected at least 2 parts in INFO entry: ID=RPPR,Number=1,Type=Float,Description="Read Placemen
Warning: Expected at least 2 parts in INFO entry: ID=EPP,Number=A,Type=Float,Description="End Placement
Warning: Expected at least 2 parts in INFO entry: ID=EPPR,Number=1,Type=Float,Description="End Placement
Warning: Expected at least 2 parts in INFO entry: ID=TYPE,Number=A,Type=String,Description="The type of
Warning: Expected at least 2 parts in INFO entry: ID=TYPE,Number=A,Type=String,Description="The type of
```

```
Warning: Expected at least 2 parts in INFO entry: ID=TYPE,Number=A,Type=String,Description="The type of
Warning: Expected at least 2 parts in INFO entry: ID=TYPE,Number=A,Type=String,Description="The type of
Warning: Expected at least 2 parts in INFO entry: ID=TYPE,Number=A,Type=String,Description="The type of
Warning: Expected at least 2 parts in INFO entry: ID=CIGAR,Number=A,Type=String,Description="The extend
Warning: Expected at least 2 parts in FORMAT entry: ID=GQ,Number=1,Type=Float,Description="Genotype Qua
Warning: Expected at least 2 parts in FORMAT entry: ID=GL,Number=G,Type=Float,Description="Genotype Lik
Keeping individuals in 'keep' list
After filtering, kept 26 out of 195 Individuals
Outputting VCF file...
After filtering, kept 30 out of a possible 30 Sites
Run Time = 0.00 seconds

VCFtools - 0.1.17
(C) Adam Auton and Anthony Marcketta 2009

Parameters as interpreted:
    --vcf DQBex2_final.recode.vcf
    --keep SGAmhckeep.txt
    --out DQBex2_SGA
    --recode

Warning: Expected at least 2 parts in INFO entry: ID=AF,Number=A,Type=Float,Description="Estimated alle
Warning: Expected at least 2 parts in INFO entry: ID=PRO,Number=1,Type=Float,Description="Reference all
Warning: Expected at least 2 parts in INFO entry: ID=PAO,Number=A,Type=Float,Description="Alternate all
Warning: Expected at least 2 parts in INFO entry: ID=SRP,Number=1,Type=Float,Description="Strand balanc
Warning: Expected at least 2 parts in INFO entry: ID=SAP,Number=A,Type=Float,Description="Strand balanc
Warning: Expected at least 2 parts in INFO entry: ID=AB,Number=A,Type=Float,Description="Allele balance
Warning: Expected at least 2 parts in INFO entry: ID=ABP,Number=A,Type=Float,Description="Allele balanc
Warning: Expected at least 2 parts in INFO entry: ID=RPP,Number=A,Type=Float,Description="Read Placemen
Warning: Expected at least 2 parts in INFO entry: ID=RPPR,Number=1,Type=Float,Description="Read Placeme
Warning: Expected at least 2 parts in INFO entry: ID=EPP,Number=A,Type=Float,Description="End Placement
Warning: Expected at least 2 parts in INFO entry: ID=EPPR,Number=1,Type=Float,Description="End Placemen
Warning: Expected at least 2 parts in INFO entry: ID=TYPE,Number=A,Type=String,Description="The type of
Warning: Expected at least 2 parts in INFO entry: ID=TYPE,Number=A,Type=String,Description="The type of
Warning: Expected at least 2 parts in INFO entry: ID=TYPE,Number=A,Type=String,Description="The type of
Warning: Expected at least 2 parts in INFO entry: ID=TYPE,Number=A,Type=String,Description="The type of
Warning: Expected at least 2 parts in INFO entry: ID=TYPE,Number=A,Type=String,Description="The type of
Warning: Expected at least 2 parts in INFO entry: ID=CIGAR,Number=A,Type=String,Description="The extend
Warning: Expected at least 2 parts in FORMAT entry: ID=GQ,Number=1,Type=Float,Description="Genotype Qua
Warning: Expected at least 2 parts in FORMAT entry: ID=GL,Number=G,Type=Float,Description="Genotype Lik
Keeping individuals in 'keep' list
After filtering, kept 25 out of 195 Individuals
Outputting VCF file...
After filtering, kept 30 out of a possible 30 Sites
Run Time = 0.00 seconds
```

Now calculate allele frequencies for each population…

```
vcftools --vcf DQBex2_CGA12.recode.vcf --mac 1 --freq --out DQBex2_CGA12
vcftools --vcf DQBex2_CGA23.recode.vcf --mac 1 --freq --out DQBex2_CGA23
vcftools --vcf DQBex2_NGA.recode.vcf --mac 1 --freq --out DQBex2_NGA
vcftools --vcf DQBex2_SGA.recode.vcf --mac 1 --freq --out DQBex2_SGA
```

```
VCFtools - 0.1.17
(C) Adam Auton and Anthony Marcketta 2009

Parameters as interpreted:
    --vcf DQBex2_CGA12.recode.vcf
    --mac 1
    --freq
    --out DQBex2_CGA12

Warning: Expected at least 2 parts in INFO entry: ID=AF,Number=A,Type=Float,Description="Estimated allel
Warning: Expected at least 2 parts in INFO entry: ID=PRO,Number=1,Type=Float,Description="Reference allo
Warning: Expected at least 2 parts in INFO entry: ID=PAO,Number=A,Type=Float,Description="Alternate allo
Warning: Expected at least 2 parts in INFO entry: ID=SRP,Number=1,Type=Float,Description="Strand balanc
Warning: Expected at least 2 parts in INFO entry: ID=SAP,Number=A,Type=Float,Description="Strand balanc
Warning: Expected at least 2 parts in INFO entry: ID=AB,Number=A,Type=Float,Description="Allele balance
Warning: Expected at least 2 parts in INFO entry: ID=ABP,Number=A,Type=Float,Description="Allele balanc
Warning: Expected at least 2 parts in INFO entry: ID=RPP,Number=A,Type=Float,Description="Read Placemen
Warning: Expected at least 2 parts in INFO entry: ID=RPPR,Number=1,Type=Float,Description="Read Placeme
Warning: Expected at least 2 parts in INFO entry: ID=EPP,Number=A,Type=Float,Description="End Placement
Warning: Expected at least 2 parts in INFO entry: ID=EPPR,Number=1,Type=Float,Description="End Placemen
Warning: Expected at least 2 parts in INFO entry: ID=TYPE,Number=A,Type=String,Description="The type of
Warning: Expected at least 2 parts in INFO entry: ID=TYPE,Number=A,Type=String,Description="The type of
Warning: Expected at least 2 parts in INFO entry: ID=TYPE,Number=A,Type=String,Description="The type of
Warning: Expected at least 2 parts in INFO entry: ID=TYPE,Number=A,Type=String,Description="The type of
Warning: Expected at least 2 parts in INFO entry: ID=TYPE,Number=A,Type=String,Description="The type of
Warning: Expected at least 2 parts in INFO entry: ID=CIGAR,Number=A,Type=String,Description="The extend
Warning: Expected at least 2 parts in FORMAT entry: ID=GQ,Number=1,Type=Float,Description="Genotype Qua
Warning: Expected at least 2 parts in FORMAT entry: ID=GL,Number=G,Type=Float,Description="Genotype Lik
After filtering, kept 12 out of 12 Individuals
Outputting Frequency Statistics...
After filtering, kept 16 out of a possible 30 Sites
Run Time = 0.00 seconds

VCFtools - 0.1.17
(C) Adam Auton and Anthony Marcketta 2009

Parameters as interpreted:
    --vcf DQBex2_CGA23.recode.vcf
    --mac 1
    --freq
    --out DQBex2_CGA23

Warning: Expected at least 2 parts in INFO entry: ID=AF,Number=A,Type=Float,Description="Estimated allel
Warning: Expected at least 2 parts in INFO entry: ID=PRO,Number=1,Type=Float,Description="Reference allo
Warning: Expected at least 2 parts in INFO entry: ID=PAO,Number=A,Type=Float,Description="Alternate allo
Warning: Expected at least 2 parts in INFO entry: ID=SRP,Number=1,Type=Float,Description="Strand balanc
Warning: Expected at least 2 parts in INFO entry: ID=SAP,Number=A,Type=Float,Description="Strand balanc
Warning: Expected at least 2 parts in INFO entry: ID=AB,Number=A,Type=Float,Description="Allele balance
Warning: Expected at least 2 parts in INFO entry: ID=ABP,Number=A,Type=Float,Description="Allele balanc
Warning: Expected at least 2 parts in INFO entry: ID=RPP,Number=A,Type=Float,Description="Read Placemen
Warning: Expected at least 2 parts in INFO entry: ID=RPPR,Number=1,Type=Float,Description="Read Placeme
Warning: Expected at least 2 parts in INFO entry: ID=EPP,Number=A,Type=Float,Description="End Placement
Warning: Expected at least 2 parts in INFO entry: ID=EPPR,Number=1,Type=Float,Description="End Placemen
Warning: Expected at least 2 parts in INFO entry: ID=TYPE,Number=A,Type=String,Description="The type of
```

```
Warning: Expected at least 2 parts in INFO entry: ID=TYPE,Number=A,Type=String,Description="The type of
Warning: Expected at least 2 parts in INFO entry: ID=TYPE,Number=A,Type=String,Description="The type of
Warning: Expected at least 2 parts in INFO entry: ID=TYPE,Number=A,Type=String,Description="The type of
Warning: Expected at least 2 parts in INFO entry: ID=TYPE,Number=A,Type=String,Description="The type of
Warning: Expected at least 2 parts in INFO entry: ID=CIGAR,Number=A,Type=String,Description="The extend
Warning: Expected at least 2 parts in FORMAT entry: ID=GQ,Number=1,Type=Float,Description="Genotype Qual
Warning: Expected at least 2 parts in FORMAT entry: ID=GL,Number=G,Type=Float,Description="Genotype Lik
After filtering, kept 135 out of 135 Individuals
Outputting Frequency Statistics...
After filtering, kept 30 out of a possible 30 Sites
Run Time = 0.00 seconds

VCFtools - 0.1.17
(C) Adam Auton and Anthony Marcketta 2009

Parameters as interpreted:
    --vcf DQBex2_NGA.recode.vcf
    --mac 1
    --freq
    --out DQBex2_NGA

Warning: Expected at least 2 parts in INFO entry: ID=AF,Number=A,Type=Float,Description="Estimated alle
Warning: Expected at least 2 parts in INFO entry: ID=PRO,Number=1,Type=Float,Description="Reference all
Warning: Expected at least 2 parts in INFO entry: ID=PAO,Number=A,Type=Float,Description="Alternate all
Warning: Expected at least 2 parts in INFO entry: ID=SRP,Number=1,Type=Float,Description="Strand balanc
Warning: Expected at least 2 parts in INFO entry: ID=SAP,Number=A,Type=Float,Description="Strand balanc
Warning: Expected at least 2 parts in INFO entry: ID=AB,Number=A,Type=Float,Description="Allele balance
Warning: Expected at least 2 parts in INFO entry: ID=ABP,Number=A,Type=Float,Description="Allele balanc
Warning: Expected at least 2 parts in INFO entry: ID=RPP,Number=A,Type=Float,Description="Read Placemen
Warning: Expected at least 2 parts in INFO entry: ID=RPPR,Number=1,Type=Float,Description="Read Placeme
Warning: Expected at least 2 parts in INFO entry: ID=EPP,Number=A,Type=Float,Description="End Placement
Warning: Expected at least 2 parts in INFO entry: ID=EPPR,Number=1,Type=Float,Description="End Placemen
Warning: Expected at least 2 parts in INFO entry: ID=TYPE,Number=A,Type=String,Description="The type of
Warning: Expected at least 2 parts in INFO entry: ID=TYPE,Number=A,Type=String,Description="The type of
Warning: Expected at least 2 parts in INFO entry: ID=TYPE,Number=A,Type=String,Description="The type of
Warning: Expected at least 2 parts in INFO entry: ID=TYPE,Number=A,Type=String,Description="The type of
Warning: Expected at least 2 parts in INFO entry: ID=TYPE,Number=A,Type=String,Description="The type of
Warning: Expected at least 2 parts in INFO entry: ID=CIGAR,Number=A,Type=String,Description="The extend
Warning: Expected at least 2 parts in FORMAT entry: ID=GQ,Number=1,Type=Float,Description="Genotype Qual
Warning: Expected at least 2 parts in FORMAT entry: ID=GL,Number=G,Type=Float,Description="Genotype Lik
After filtering, kept 26 out of 26 Individuals
Outputting Frequency Statistics...
After filtering, kept 23 out of a possible 30 Sites
Run Time = 0.00 seconds

VCFtools - 0.1.17
(C) Adam Auton and Anthony Marcketta 2009

Parameters as interpreted:
    --vcf DQBex2_SGA.recode.vcf
    --mac 1
    --freq
    --out DQBex2_SGA
```

```
Warning: Expected at least 2 parts in INFO entry: ID=AF,Number=A,Type=Float,Description="Estimated allel
Warning: Expected at least 2 parts in INFO entry: ID=PRO,Number=1,Type=Float,Description="Reference all
Warning: Expected at least 2 parts in INFO entry: ID=PAO,Number=A,Type=Float,Description="Alternate all
Warning: Expected at least 2 parts in INFO entry: ID=SRP,Number=1,Type=Float,Description="Strand balanc
Warning: Expected at least 2 parts in INFO entry: ID=SAP,Number=A,Type=Float,Description="Strand balanc
Warning: Expected at least 2 parts in INFO entry: ID=AB,Number=A,Type=Float,Description="Allele balance
Warning: Expected at least 2 parts in INFO entry: ID=ABP,Number=A,Type=Float,Description="Allele balanc
Warning: Expected at least 2 parts in INFO entry: ID=RPP,Number=A,Type=Float,Description="Read Placemen
Warning: Expected at least 2 parts in INFO entry: ID=RPPR,Number=1,Type=Float,Description="Read Placeme
Warning: Expected at least 2 parts in INFO entry: ID=EPP,Number=A,Type=Float,Description="End Placement
Warning: Expected at least 2 parts in INFO entry: ID=EPPR,Number=1,Type=Float,Description="End Placemen
Warning: Expected at least 2 parts in INFO entry: ID=TYPE,Number=A,Type=String,Description="The type of
Warning: Expected at least 2 parts in INFO entry: ID=TYPE,Number=A,Type=String,Description="The type of
Warning: Expected at least 2 parts in INFO entry: ID=TYPE,Number=A,Type=String,Description="The type of
Warning: Expected at least 2 parts in INFO entry: ID=TYPE,Number=A,Type=String,Description="The type of
Warning: Expected at least 2 parts in INFO entry: ID=TYPE,Number=A,Type=String,Description="The type of
Warning: Expected at least 2 parts in INFO entry: ID=CIGAR,Number=A,Type=String,Description="The extend
Warning: Expected at least 2 parts in FORMAT entry: ID=GQ,Number=1,Type=Float,Description="Genotype Qua
Warning: Expected at least 2 parts in FORMAT entry: ID=GL,Number=G,Type=Float,Description="Genotype Lik
After filtering, kept 25 out of 25 Individuals
Outputting Frequency Statistics...
After filtering, kept 20 out of a possible 30 Sites
Run Time = 0.00 seconds
```

Read in the frequency info and annotation info…

```r
#Read the annotation information and files with info on variable positions
anns <- read.csv("DQBex2ann.csv", header=TRUE)
CGA12frq <- read.table("DQBex2_CGA12.frq", header=FALSE, skip=1)
colnames(CGA12frq) <- c("CHROM","POS","N_ALLELES","A1","A2")
CGA23frq <- read.table("DQBex2_CGA23.frq", header=FALSE, skip=1)
colnames(CGA23frq) <- c("CHROM","POS","N_ALLELES","A1","A2")
NGAfrq <- read.table("DQBex2_NGA.frq", header=FALSE, skip=1)
colnames(NGAfrq) <- c("CHROM","POS","N_ALLELES","A1","A2")
SGAfrq <- read.table("DQBex2_SGA.frq", header=FALSE, skip=1)
colnames(SGAfrq) <- c("CHROM","POS","N_ALLELES","A1","A2")

#Extract polymorphic positions for each pop
CGA12anns <- anns[which(anns$POS %in% CGA12frq$POS),3]
CGA23anns <- anns[which(anns$POS %in% CGA23frq$POS),3]
NGAanns <- anns[which(anns$POS %in% NGAfrq$POS),3]
SGAanns <- anns[which(anns$POS %in% SGAfrq$POS),3]

#Table with mutation types
table(sapply(CGA12anns, FUN=function(x) strsplit(x, "|", fixed=TRUE)[[1]][2]))
```

```
##
##   missense_variant synonymous_variant
##                 14                  2
```

```r
table(sapply(CGA23anns, FUN=function(x) strsplit(x, "|", fixed=TRUE)[[1]][2]))
```

```
##
```

```
##   missense_variant synonymous_variant
##                 24                  6
```

```r
table(sapply(NGAanns, FUN=function(x) strsplit(x, "|", fixed=TRUE)[[1]][2]))
```

```
##
##   missense_variant synonymous_variant
##                 19                  4
```

```r
table(sapply(SGAanns, FUN=function(x) strsplit(x, "|", fixed=TRUE)[[1]][2]))
```

```
##
##   missense_variant synonymous_variant
##                 18                  2
```

Summary

Haplotype diversity and observed heterozygosity are both fairly high in central GA. Overall, central GA has retained a surprising amount of diversity in exon 2 of *DQB*. Furthermore, the haplotype network and neighbor-joining tree show that variants segregating in the central GA population are not particularly tightly clustered. However, many of the variants in central GA are present at low frequencies.

```r
knitr::kable(t(out), digits = c(0,0,3,3,0))
```

	n	K_hap	H_hap	Ho_hap	missense
NGA	26	19	0.870	0.615	19
CGA12	12	7	0.848	0.417	14
CGA23	135	30	0.862	0.659	24
SGA	25	14	0.861	0.640	18
